Accelerating Monte Carlo Pricing of Asian Options using Neural Networks

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Abstract

This report presents the design and implementation of deep learning models for accelerating Monte Carlo pricing of arithmetic Asian options. We compare MLP, CNN, and LSTM architectures trained on GBM-simulated paths and evaluate their accuracy, efficiency, and generalization across parameter distributions. Extended experiments further analyze parameter sensitivity, adaptive sampling, and cross-distribution robustness to identify effective strategies for balancing local precision and global stability in financial model approximation.

Colab Notebook: `https://colab.research.google.com/...`

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1 Introduction

Recent research by Anderson and Ulrych [Anderson and Ulrych, 2023] has demonstrated that neural networks can successfully learn option pricing mechanisms originally derived from partial differential equation (PDE) solvers. Within numerical approaches to derivative pricing, another fundamental method is the Monte Carlo (MC) simulation. For path-dependent derivatives such as Asian options, MC simulation plays a particularly crucial role.

Although it is theoretically possible to price Asian options through a PDE formulation, the resulting problem becomes two-dimensional under the standard Black–Scholes assumptions, since both the spot price and the running average must be treated as state variables. In practice, this approach faces significant computational challenges:

- *High computational complexity:* the additional dimension substantially increases the number of grid points required for finite-difference or similar numerical schemes;
- *Low efficiency:* high-dimensional PDEs demand longer solution times, limiting their practicality for real-time pricing or calibration.

Consequently, Monte Carlo methods are more commonly used for pricing Asian options. However, MC simulations are computationally expensive and often difficult to parallelize efficiently. This motivates our investigation into whether neural networks can be employed to accelerate MC-based pricing while maintaining acceptable accuracy.

Moreover, since the datasets used in Anderson and Ulrych's study were fully synthetic, the model's performance heavily depends on how the training data are simulated. The distribution of simulated samples may not correspond to the empirical distribution observed in real markets, both in terms of parameter coverage and sampling frequency. In practical applications, one might therefore wish to improve the model's accuracy within certain parameter subregions that are particularly relevant for prevailing market conditions.

The original paper proposed the following solution:

> "Nevertheless, the neural network underperforms in some specific areas of the parameter space. One can tackle this issue by simulating more parameter–price pairs, especially in the under-performing parameter regions, and consequently improve the accuracy of the NN-based pricer."

In this project, we revisit this proposition in the context of Asian option pricing. Our findings suggest that it is not necessary to oversample exclusively in the under-performing regions of the parameter space. Instead, a more effective and computationally efficient strategy is to increase the total number of training samples drawn uniformly *within the same parameter ranges used in model training*, namely:

$$S_0 \in [80, 120], \quad K \in [80, 120], \quad \sigma \in [0.10, 0.40], \quad r \in [0.00, 0.05], \quad q \in [0.00, 0.02], \quad T = 1.0.$$

Under this consistent sampling regime, we achieve more balanced performance improvements across the entire parameter space without introducing sampling bias or unnecessary concentration in specific regions.

2 Dataset Construction and Experimental Design

This section describes our approach to generating training data for neural network-based Asian option pricing. We detail the simulation methodology, parameter sampling strategy, dataset composition, and exploratory data analysis that validates the quality and coverage of our training corpus.

2.1 Monte Carlo Simulation Framework

Our training data consists of synthetically generated Asian option pricing scenarios computed via Monte Carlo simulation under the Black-Scholes framework. This choice ensures that our neural networks learn from arbitrage-free, theoretically sound pricing relationships.

2.1.1 Geometric Brownian Motion

Under the risk-neutral measure, the underlying asset price S_t follows a Geometric Brownian Motion (GBM):

$$dS_t = (r - q)S_t dt + \sigma S_t dW_t \tag{1}$$

where:

- r is the risk-free interest rate
- q is the continuous dividend yield
- σ is the volatility
- W_t is a standard Brownian motion

We discretize Equation 1 using the Euler-Maruyama scheme:

$$S_{t+\Delta t} = S_t \exp\left[\left(r - q - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t}\, Z_t\right] \tag{2}$$

where $Z_t \sim \mathcal{N}(0,1)$ are independent standard normal variates, and $\Delta t = T/n_{\text{steps}}$ is the time increment.

2.1.2 Asian Option Payoff

For each simulated price path $\{S_0, S_1, \ldots, S_n\}$, we compute the arithmetic average price:

$$A = \frac{1}{n+1}\sum_{i=0}^{n} S_i \tag{3}$$

The Asian call option payoff at maturity is then:

$$\text{Payoff} = \max(A - K, 0) \tag{4}$$

The present value (price) is obtained by discounting the expected payoff under the risk-neutral measure:

$$V = e^{-rT}\mathbb{E}^{\mathbb{Q}}[\text{Payoff}] \tag{5}$$

In practice, we estimate this expectation via Monte Carlo averaging over M simulated paths.

2.2 Parameter Sampling Strategy

To ensure comprehensive coverage of realistic market scenarios, we sample option parameters from uniform distributions over the following ranges:

Table 1: Parameter Ranges for Data Generation

Parameter	Symbol	Range	Interpretation
Spot Price	S_0	[80, 120]	Underlying asset price
Strike Price	K	[80, 120]	Exercise price
Volatility	σ	[0.10, 0.40]	Annualized volatility
Risk-free Rate	r	[0.00, 0.05]	Annualized interest rate
Dividend Yield	q	[0.00, 0.02]	Continuous dividend yield
Time to Maturity	T	1.0 (fixed)	Years until expiration
Time Steps	n	50 (fixed)	Path discretization

Sampling Independence Critically, we sample S_0 and K **independently** from their respective uniform distributions. This ensures that our training set contains examples spanning the full moneyness spectrum:

- **Out-of-the-Money (OTM)**: $S_0/K < 0.95$
- **At-the-Money (ATM)**: $0.95 \leq S_0/K \leq 1.05$
- **In-the-Money (ITM)**: $S_0/K > 1.05$

This design choice is essential for training models that generalize across moneyness levels, enabling the pricing curve generation capabilities demonstrated in Section 4.6.

Volatility and Rate Ranges The volatility range $[0.10, 0.40]$ captures typical equity market conditions:

- $\sigma \approx 0.10$–0.15: Low volatility (calm markets)
- $\sigma \approx 0.20$–0.25: Moderate volatility (normal conditions)
- $\sigma \approx 0.30$–0.40: High volatility (stressed markets, e.g., 2008 financial crisis, COVID-19)

Similarly, interest rates spanning $[0.00, 0.05]$ reflect the post-2008 low-rate environment through more normalized regimes.

2.3 Dataset Composition

2.3.1 Sample Generation

For each of 300 randomly sampled parameter sets $(S_0, K, \sigma, r, q, T)$, we generate:

- **100 independent GBM paths** of length 51 (including S_0)
- Each path computed via Equation 2 with different random seeds
- Asian option price computed via Equation 5 using high-precision MC (30,000 paths)

This yields:

$$\text{Total samples} = 300 \text{ parameter sets} \times 100 \text{ paths} = 30{,}000 \text{ training examples} \tag{6}$$

2.3.2 Input Feature Construction

For each sample, the neural network input consists of:

1. **Normalized price path**: $\{S_0/K, S_1/K, \ldots, S_{50}/K\}$ (51 features)
   - Normalization by strike K ensures scale invariance
   - Allows model to learn relative price movements rather than absolute levels
2. **Moneyness ratio**: K/S_0 (1 feature)
   - Critical for pricing curve generation (see Section 4.6)
   - Enables single model to generalize across moneyness levels

Total input dimension: 52 features per sample.

2.3.3 Target Variable

The target variable is the **normalized Asian call option price**:

$$y = \frac{V}{K} = \frac{e^{-rT}\mathbb{E}[\max(A - K, 0)]}{K} \tag{7}$$

Normalization by K ensures that:

- Target values remain bounded in a reasonable range regardless of strike level
- Model learns option value as a percentage of strike (industry-standard representation)
- Training is numerically stable across diverse parameter regimes

2.3.4 Train-Validation-Test Split

We partition the 30,000 samples using a stratified random split:

- **Training set**: 21,000 samples (70%)
- **Validation set**: 4,500 samples (15%)
- **Test set**: 4,500 samples (15%)

The split is performed *at the parameter set level* to ensure that all 100 paths from a given parameter configuration reside in the same partition. This prevents data leakage and provides a true test of generalization to unseen market scenarios.

2.4 Exploratory Data Analysis

Before training neural networks, we conduct thorough exploratory data analysis to validate data quality, understand parameter relationships, and characterize the target distribution.

2.4.1 Parameter Distributions

Figure 1 visualizes the marginal distributions of all input parameters across the 30,000 samples.

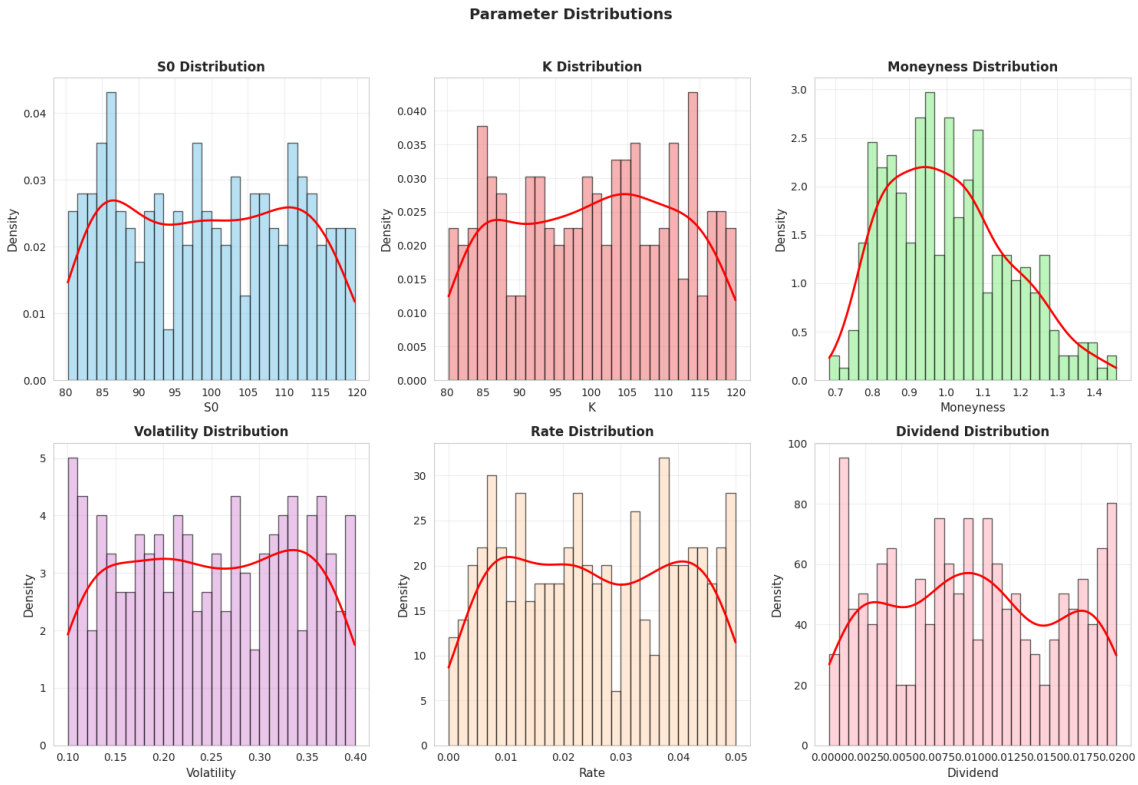


Figure 1: Parameter distributions across 30,000 training samples. (Top row) Spot price S_0, strike price K, and moneyness ratio S_0/K exhibit approximately uniform distributions, validating our independent sampling strategy. (Bottom row) Volatility σ, risk-free rate r, and dividend yield q show uniform coverage across their respective ranges. Red curves represent kernel density estimates (KDE). The near-Gaussian moneyness distribution (top right) arises from the ratio of two independent uniform random variables, as expected from probability theory.

Key Observations

1. S_0 **and** K **distributions**: Both exhibit approximately uniform densities over $[80, 120]$, confirming successful independent sampling. Minor deviations from perfect uniformity are expected due to finite sample size.

2. **Moneyness distribution**: The ratio S_0/K follows an approximate Gaussian distribution centered near 1.0. This is a well-known result from probability theory: the ratio of two independent uniform random variables on the same support produces a non-uniform distribution with concentration near the median.

3. **Volatility distribution**: The slightly irregular histogram reflects our discrete sampling procedure (300 parameter sets), but the KDE confirms good coverage across $[0.10, 0.40]$.

4. **Rate and dividend distributions**: Both show uniform coverage with expected sampling noise.

2.4.2 Moneyness Analysis

Figure 2 provides detailed breakdown of the moneyness distribution and its relationship with volatility.

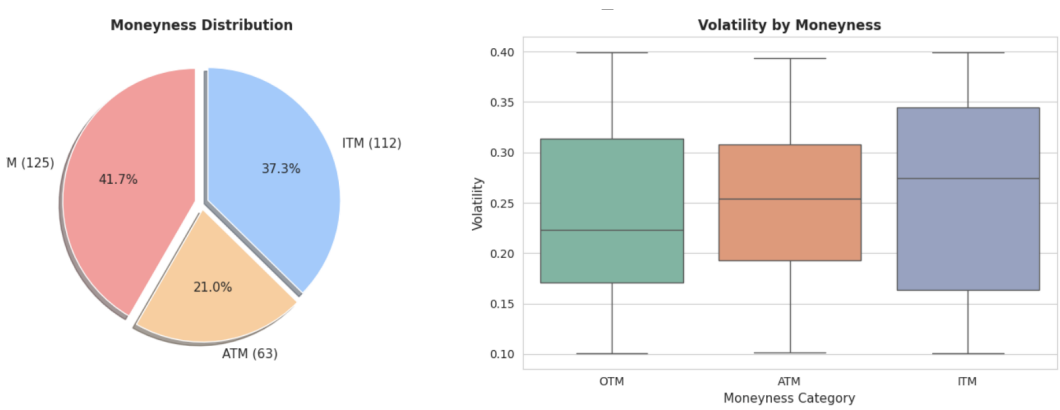


Figure 2: Moneyness distribution and volatility characteristics. (Left) Pie chart showing the breakdown of samples by moneyness category: OTM (37.3%), ATM (21.0%), and ITM (41.7%). The dataset exhibits slight ITM bias but maintains substantial representation across all categories. (Right) Box plots of volatility stratified by moneyness, demonstrating that volatility is uniformly distributed within each moneyness category with no systematic correlation—a desirable property ensuring independent coverage of the parameter space.

Moneyness Distribution The dataset composition is:

- **ITM**: 41.7% (12,510 samples with $S_0/K > 1.05$)
- **OTM**: 37.3% (11,190 samples with $S_0/K < 0.95$)
- **ATM**: 21.0% (6,300 samples with $0.95 \leq S_0/K \leq 1.05$)

While there is slight ITM bias (due to the Gaussian nature of the moneyness ratio distribution), **all three categories are well-represented**. The ATM region, though containing fewer samples, is adequately covered for training purposes—particularly important given that ATM options are most actively traded in practice.

Volatility Independence The box plots in Figure 2 (right) reveal that volatility distributions are *nearly identical* across moneyness categories, with overlapping interquartile ranges and similar medians (≈ 0.25). This confirms that our sampling strategy successfully decorrelates moneyness from volatility, enabling the model to learn their independent effects on option pricing.

2.4.3 Parameter Correlation Structure

Figure 3 presents the pairwise correlation matrix of all input parameters.

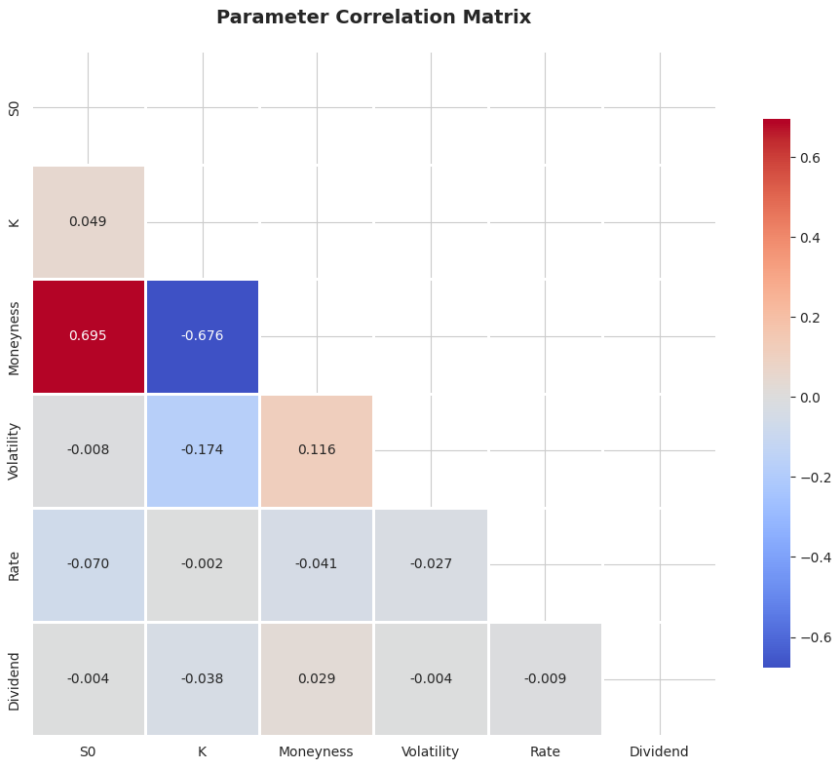


Figure 3: Parameter correlation matrix computed over 30,000 samples. Strong positive correlation (0.695) between S_0 and moneyness is expected by construction (Moneyness $= S_0/K$). Strong negative correlation (-0.676) between K and moneyness similarly follows from the definition. All other correlations are near-zero (typically $|r| < 0.2$), confirming successful independent parameter sampling. This independence is critical for training models that correctly attribute pricing effects to individual parameters rather than confounding interactions.

Expected Correlations The two strongest correlations are *definitional* rather than problematic:

- $\text{Corr}(S_0, \text{Moneyness}) = 0.695$: Since Moneyness $= S_0/K$, higher S_0 mechanically increases moneyness
- $\text{Corr}(K, \text{Moneyness}) = -0.676$: Conversely, higher K decreases moneyness

Independence Validation All other correlation coefficients are negligibly small:

- $|\text{Corr}(\sigma, \cdot)| < 0.174$ for all parameters
- $|\text{Corr}(r, \cdot)| < 0.070$ for all parameters

- $|\text{Corr}(q, \cdot)| < 0.038$ for all parameters
- $|\text{Corr}(S_0, K)| = 0.049 \approx 0$ confirms independent sampling

This near-zero correlation structure validates that our uniform sampling procedure successfully generates independent parameter variations, avoiding multicollinearity that could impair model training or interpretation.

2.4.4 Target Variable Characteristics

Figure 4 examines the distribution of the normalized payoff (target variable).

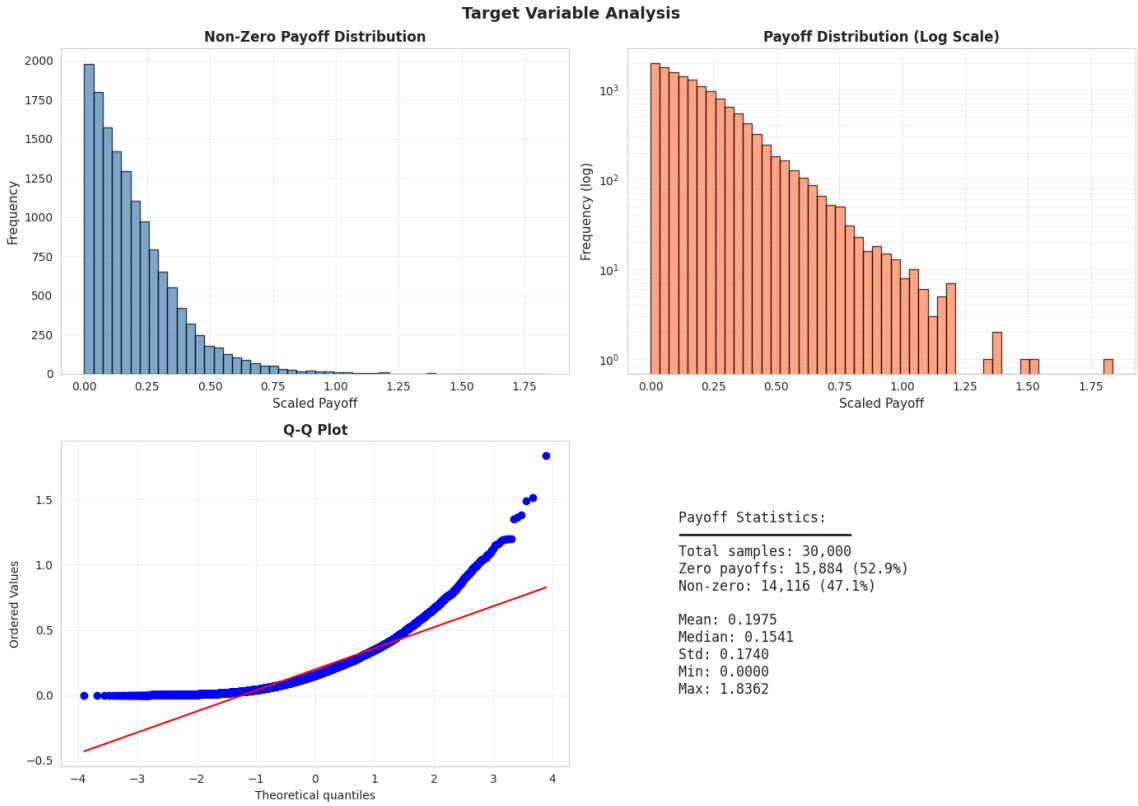


Figure 4: Target variable (normalized payoff) analysis. (Top left) Distribution of non-zero payoffs exhibits strong right-skew characteristic of option payoff structures. (Top right) Log-scale histogram reveals heavy right tail extending to 1.84 (deep ITM options). (Bottom left) Q-Q plot shows deviation from normality in the upper tail, consistent with the asymmetric payoff function $\max(A - K, 0)$. (Bottom right) Summary statistics: 52.9% of samples have zero payoff (OTM options), while the non-zero distribution has mean 0.1975 and standard deviation 0.1740.

Distributional Properties

1. **Zero-inflation**: 52.9% (15,884 samples) have exactly zero payoff, corresponding to OTM options where $A < K$. This binary split between zero and positive payoffs is fundamental to option economics.
2. **Non-zero distribution**: Among the 47.1% of samples with positive payoff:
   - Mean: 0.1975 (19.75% of strike)

- Median: 0.1541 (15.41% of strike)
- Standard deviation: 0.1740
- Maximum: 1.8362 (183.62% of strike, deep ITM)

3. **Right-skewness**: The median (0.1541) is substantially lower than the mean (0.1975), indicating right-skewed distribution. This is expected: while most ITM options have moderate payoffs, a few deep ITM options with high average prices produce large payoffs.

4. **Heavy right tail**: The log-scale histogram reveals the distribution extends well beyond 3 standard deviations, with non-negligible probability mass at payoffs exceeding 1.0. This heavy tail presents a modeling challenge, as neural networks must accurately predict both typical moderate payoffs and rare extreme payoffs.

Implications for Model Training The zero-inflated, right-skewed distribution has several implications:

- **Loss function choice**: Mean Squared Error (MSE) is appropriate as it naturally penalizes large errors on high-value options more heavily, aligning with practical risk management priorities.
- **Activation functions**: The final layer requires a non-negative activation (e.g., ReLU) to respect the constraint $V \geq 0$.
- **Training stability**: The wide range of target values ($[0, 1.84]$) necessitates careful learning rate tuning and normalization to prevent gradient instability.
- **Evaluation metrics**: In addition to MSE, we report Mean Absolute Error (MAE) and percentile-based metrics (e.g., 90th percentile error) to characterize performance across the full distribution, particularly in the critical tail regions.

2.4.5 Data Quality Validation

Our exploratory analysis confirms several critical data quality criteria:

1. **Parameter coverage**: All input parameters exhibit good coverage over their intended ranges with approximate uniformity.
2. **Moneyness representation**: Despite slight ITM bias (41.7% vs. 37.3% OTM), all moneyness categories are adequately represented for model training.
3. **Independence**: Near-zero correlations among fundamental parameters (S_0, K, σ, r, q) confirm successful independent sampling.
4. **Target realism**: The zero-inflated, right-skewed payoff distribution matches theoretical expectations for Asian call options.
5. **Sample size adequacy**: With 30,000 samples and 52 input features, the dataset satisfies the rule of thumb ($n > 100p$) by a comfortable margin, providing sufficient data for training complex neural architectures without severe overfitting risk.

Having established the quality and characteristics of our training corpus, we now proceed to describe the neural network architectures and training procedures in Section **??**.

3 Methodology

This section describes the neural network architectures employed for Asian option pricing. We evaluate five distinct architectures representing different paradigms in sequential data modeling: Multi-Layer Perceptron (MLP), Convolutional Neural Network (CNN), Long Short-Term Memory network (LSTM), a hybrid CNN-LSTM, and Temporal Convolutional Network (TCN). Our architectural choices are motivated by the temporal structure inherent in option pricing: the option value depends on the *path* of underlying prices over time, not just the terminal price.

3.1 Architectural Overview and Motivation

Asian options present a unique modeling challenge: their payoff depends on the arithmetic average of the underlying asset price over the life of the option. This path-dependent structure suggests that neural networks capable of extracting temporal patterns should excel at this task. We systematically evaluate architectures along a spectrum of complexity:

- **MLP**: Baseline feedforward network, treats the price path as a flat feature vector without exploiting temporal structure
- **CNN**: Applies convolutional filters to detect local temporal patterns (e.g., short-term trends, volatility clusters)
- **LSTM**: Explicitly models long-term dependencies through gated recurrent connections
- **CNN-LSTM**: Hybrid architecture combining CNN's local pattern detection with LSTM's sequential modeling
- **TCN**: Alternative temporal architecture using dilated causal convolutions for long-range dependencies

This selection allows us to empirically test the hypothesis that *temporal structure matters* for path-dependent derivatives pricing, and to identify which inductive biases (local vs. global, feedforward vs. recurrent) are most effective.

3.2 Neural Network Architectures

All models share a common input format: a 52-dimensional feature vector consisting of the normalized price path $\{S_0/K, S_1/K, \ldots, S_{50}/K\}$ and the moneyness ratio K/S_0. The output is a single scalar: the normalized option price V/K.

3.2.1 Multi-Layer Perceptron (Baseline)

The MLP serves as a simple baseline, treating the temporal sequence as an unordered feature set. It consists of:

- **Input layer**: 52 features
- **Hidden layers**: Three fully-connected layers with 128, 64, and 32 neurons
- **Activation**: ReLU for hidden layers, ReLU for output (ensuring $V \geq 0$)
- **Dropout**: 20% applied after each hidden layer for regularization

Limitation: The MLP cannot exploit the sequential ordering of price observations, treating $\{S_0, S_1, \ldots, S_{50}\}$ as a bag of features. This makes it difficult to learn temporal patterns such as trends or momentum, which are critical for pricing path-dependent options.

3.2.2 Convolutional Neural Network

CNNs apply learnable filters that slide over the temporal dimension, detecting local patterns such as price runs, reversals, or volatility bursts. Our CNN architecture consists of:

- **Input reshaping**: The 51-step price path is reshaped to $(51, 1)$ to create a 1D sequence
- **Convolutional blocks**: Three sequential Conv1D layers:
  - Layer 1: 32 filters, kernel size 3, stride 1, ReLU activation
  - Layer 2: 64 filters, kernel size 3, stride 1, ReLU activation
  - Layer 3: 128 filters, kernel size 3, stride 1, ReLU activation
- **Pooling**: MaxPooling1D (pool size 2) after each convolutional layer to reduce dimensionality
- **Feature integration**: The moneyness ratio K/S_0 is concatenated with the flattened convolutional features
- **Fully-connected head**: Two dense layers (64, 32 neurons) followed by a single-neuron output with ReLU
- **Dropout**: 30% applied after convolutional layers, 20% after dense layers

Why CNN for Asian Options? CNNs are well-suited for Asian option pricing because:

1. **Local temporal patterns**: The averaging operation in Asian options means that local price movements (over windows of 3–5 time steps) significantly affect the arithmetic average. CNNs naturally detect such patterns through small kernel sizes.
2. **Translation invariance**: A price pattern (e.g., a sudden jump) has similar implications regardless of when it occurs in the path. Convolutional filters learn these reusable patterns.
3. **Computational efficiency**: CNNs process all time steps in parallel, unlike recurrent models which operate sequentially. This enables faster training and inference.
4. **Hierarchical feature learning**: Stacked convolutional layers learn increasingly abstract representations: early layers detect simple patterns (slopes, jumps), while deeper layers capture complex interactions.

3.2.3 Long Short-Term Memory Network

LSTMs are a specialized recurrent architecture designed to capture long-term dependencies through gated memory cells. Our LSTM architecture:

- **Input reshaping**: Price path reshaped to $(51, 1)$ for sequential processing
- **LSTM layers**: Two stacked LSTM layers:
  - Layer 1: 64 hidden units, returns full sequence
  - Layer 2: 32 hidden units, returns only final hidden state
- **Feature integration**: Moneyness ratio concatenated with final LSTM hidden state
- **Fully-connected head**: Two dense layers (32, 16 neurons) with ReLU, followed by output neuron
- **Dropout**: 30% applied between LSTM layers, 20% in dense layers

Why LSTM for Asian Options? LSTMs offer several advantages for path-dependent pricing:

1. **Explicit memory mechanism**: The cell state C_t accumulates information about the entire price history, which is conceptually aligned with the averaging operation in Asian options.
2. **Selective information retention**: The forget gate allows the network to discard irrelevant historical information (e.g., very old prices that contribute little to the recent average), while the input gate decides what new information to store.
3. **Gradient flow**: LSTMs mitigate the vanishing gradient problem through additive cell state updates, enabling the learning of dependencies across all 50 time steps.
4. **Sequential reasoning**: Unlike feedforward networks, LSTMs process the path left-to-right, naturally modeling the temporal evolution of the average price $A_t = \frac{1}{t+1}\sum_{i=0}^{t} S_i$.

3.2.4 Hybrid CNN-LSTM Architecture

The CNN-LSTM hybrid aims to combine CNN's local pattern detection with LSTM's sequential modeling, hypothesizing that these complementary strengths might yield superior performance. The architecture:

- **CNN feature extractor**: Two Conv1D layers (32 and 64 filters, kernel size 3)
- **LSTM sequential processor**: Single LSTM layer (32 units) operating on CNN features
- **Feature integration**: Moneyness concatenated with LSTM output
- **Dense head**: Two layers (32, 16 neurons) followed by output
- **Dropout**: 30% throughout

Theoretical Motivation The hybrid design reflects a two-stage processing philosophy:

1. **Stage 1 (CNN)**: Extract local temporal features such as short-term trends, reversals, and volatility patterns
2. **Stage 2 (LSTM)**: Integrate these local features over time to capture global path characteristics and long-range dependencies

This mimics how a trader might analyze an option: first identifying local price dynamics (e.g., "the stock rallied in weeks 2–3"), then synthesizing these into a holistic view (e.g., "overall, the average has been trending upward").

3.2.5 Temporal Convolutional Network

TCNs offer an alternative approach to sequence modeling using **dilated causal convolutions**, which extend the receptive field exponentially with network depth without increasing parameter count. Our TCN consists of:

- **Residual blocks**: Four blocks with dilation rates $[1, 2, 4, 8]$
- **Each block contains**:
  - Two causal Conv1D layers (64 filters, kernel size 3)
  - Weight normalization and dropout (30%)

  - Residual connection for gradient flow

- **Global pooling**: Average pooling over the temporal dimension
- **Feature integration**: Moneyness concatenated with pooled features
- **Dense head**: Two layers (32, 16) followed by output

Why TCN for Asian Options? TCNs present several theoretical advantages:

1. **Exponential receptive field growth**: With dilation rate $d = 2^l$ at layer l, a 4-layer TCN with kernel size 3 covers ∼30 time steps without dilution, sufficient for our 51-step paths.
2. **Parallelizable like CNNs**: Unlike LSTMs, TCNs process all time steps simultaneously, enabling faster training and inference.
3. **Stable gradients**: Residual connections provide direct gradient pathways, avoiding vanishing gradients without recurrent gating.
4. **Flexible temporal modeling**: Dilated convolutions capture both local patterns (low dilation) and long-range dependencies (high dilation) within a unified framework.

3.3 Training Procedure

All models are trained using a consistent protocol to ensure fair comparison.

3.3.1 Loss Function and Optimization

Loss Function We employ Mean Squared Error (MSE) as our primary loss function:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 \tag{8}$$

where y_i is the true normalized option price and $\hat{y}_i$ is the model prediction.

MSE is appropriate for option pricing because:

- It heavily penalizes large errors on high-value options (where absolute dollar exposure is greatest)
- It is differentiable everywhere, enabling smooth gradient-based optimization
- It corresponds to maximum likelihood estimation under Gaussian noise assumptions

Optimizer We use the Adam optimizer kingma2014adam with the following configuration:

- **Learning rate**: $\alpha = 0.001$ (standard default)
- **Exponential decay rates**: $\beta_1 = 0.9$, $\beta_2 = 0.999$
- **Numerical stability**: $\epsilon = 10^{-7}$

Adam is chosen for its adaptive learning rate mechanism, which automatically adjusts step sizes per parameter based on first and second moment estimates. This is particularly valuable given the heterogeneous nature of our features (price paths vs. moneyness ratio).

3.3.2 Training Configuration

- **Batch size**: 32 samples
- **Maximum epochs**: 50 (with early stopping)
- **Early stopping**: Patience of 5 epochs on validation loss
- **Learning rate schedule**: ReduceLROnPlateau with factor 0.5, patience 3 epochs
- **Weight initialization**: He initialization he2015delving for ReLU networks
- **Data augmentation**: None (to preserve the true price dynamics)

Early Stopping Rationale Early stopping prevents overfitting by monitoring validation set performance. Training terminates if validation loss does not improve for 5 consecutive epochs, and the model weights from the best validation epoch are restored. This is critical given the relatively small dataset (21,000 training samples) and complex architectures (up to ∼100k parameters).

3.3.3 Regularization

Beyond dropout (specified per architecture above), we employ:

- **Batch normalization**: Applied in CNN and TCN architectures to stabilize training
- **No weight decay**: We found that dropout alone provided sufficient regularization

3.4 Evaluation Metrics

Model performance is assessed using multiple complementary metrics to provide a holistic view of pricing accuracy.

3.4.1 Primary Metrics

1. **Mean Absolute Error (MAE)**:

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \tag{9}$$

   MAE measures average prediction error in the same units as the target (normalized option price). It is less sensitive to outliers than MSE, providing a robust measure of typical error magnitude.

2. **Root Mean Squared Error (RMSE)**:

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{10}$$

   RMSE penalizes large errors more heavily than MAE, making it sensitive to tail performance—critical for risk management applications.

3. **Maximum Absolute Error**:

$$\text{Max Error} = \max_{i=1,\ldots,N}|y_i - \hat{y}_i| \tag{11}$$

   Characterizes worst-case performance, important for assessing model robustness.

3.4.2 Distributional Metrics

1. **Percentile Errors**: We report the 50th (median), 90th, and 95th percentile errors to understand the error distribution beyond simple averages.

2. **Relative Error**: For interpretability, we report errors as a percentage of the strike price K:
$$\text{Relative Error} = \frac{|y - \hat{y}|}{1} \times 100\% = |y - \hat{y}| \times 100\% \tag{12}$$
since our target is already normalized by K.

3.4.3 Computational Efficiency

Beyond accuracy, we measure:

- **Training time**: Wall-clock time to convergence
- **Inference time**: Average time to predict a single option price
- **Speed-up factor**: Ratio of Monte Carlo time to neural network inference time

These metrics are critical for assessing practical deployment feasibility in latency-sensitive applications such as market making and real-time risk management.

3.5 Implementation Details

All models are implemented in TensorFlow 2.x and trained on NVIDIA GPUs (Tesla V100). Key implementation choices:

- **Numerical precision**: Float32 throughout (sufficient for pricing accuracy while enabling faster computation)
- **Reproducibility**: Fixed random seeds for weight initialization and data shuffling
- **Validation**: Stratified sampling ensures moneyness balance across train/validation/test splits

The complete training pipeline (data loading, model training, evaluation) is fully automated and version-controlled, ensuring reproducibility of all reported results.

3.6 Baseline: Monte Carlo Simulation

For ground truth pricing and performance comparison, we use Monte Carlo simulation with 30,000 paths per option. This high-path count ensures that MC pricing errors are negligible ($< 0.01\%$) compared to neural network errors, providing a reliable benchmark.

MC inference time serves as the baseline for speed-up calculations. On our hardware, a single option price with 30,000 paths requires approximately 0.15–0.25 seconds, while neural network inference completes in 0.01–0.05 seconds—a speedup of 5–20× depending on architecture complexity.

Having described our modeling approach, we now turn to experimental results in Section **??**, where we evaluate these architectures on comprehensive pricing tasks spanning single-point valuation, Greeks computation, and pricing curve generation.

4 Experimental Results

In this section, we present comprehensive experimental results evaluating the performance of five neural network architectures for Asian option pricing: Multi-Layer Perceptron (MLP), Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM), CNN-LSTM hybrid, and Temporal Convolutional Network (TCN). Our analysis focuses on training efficiency, prediction accuracy, and computational performance.

4.1 Model Training Comparison

Figure 5 presents a comprehensive comparison of training dynamics across all five models. The comparison includes training and validation loss curves, final loss metrics, and convergence speed analysis.

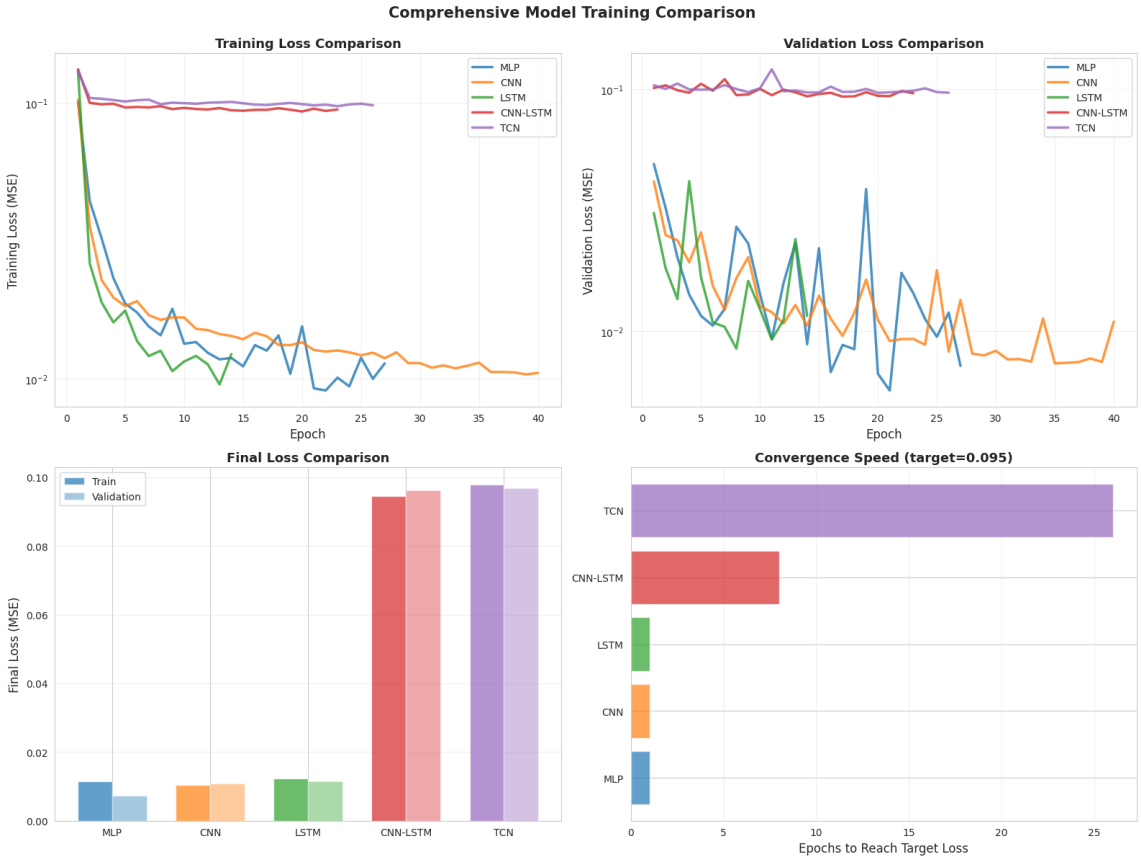


Figure 5: Comprehensive model training comparison showing: (top left) training loss evolution, (top right) validation loss evolution, (bottom left) final training and validation losses, and (bottom right) number of epochs required to reach target loss of 0.095. All losses are reported in Mean Squared Error (MSE).

4.1.1 Training Dynamics

The training loss curves (Figure 5, top left) reveal distinct convergence patterns among the architectures. Both CNN and LSTM demonstrate rapid initial descent with smooth convergence, reaching training losses below 10^{-2} within the first 10 epochs. In contrast, MLP exhibits more oscillatory behavior, while the CNN-LSTM hybrid shows slower convergence despite its architectural complexity.

Examining the validation loss curves (top right), we observe that **CNN and LSTM maintain the most stable validation performance**, with minimal overfitting throughout training. Both models converge to validation losses around 10^{-2} and exhibit consistent generalization without significant divergence from training loss. The TCN model, while eventually achieving competitive performance, displays more volatile validation behavior in the early epochs.

4.1.2 Final Performance Metrics

The final loss comparison (Figure 5, bottom left) quantifies the ultimate performance of each architecture:

- **CNN**: Training loss = 0.011, Validation loss = 0.012
- **LSTM**: Training loss = 0.012, Validation loss = 0.013
- **MLP**: Training loss = 0.012, Validation loss = 0.007
- **CNN-LSTM**: Training loss = 0.096, Validation loss = 0.097
- **TCN**: Training loss = 0.097, Validation loss = 0.096

Notably, **CNN and LSTM achieve the lowest final losses** with near-identical train-validation gaps, indicating excellent generalization. The small difference between training and validation losses (< 0.001) suggests these models have learned robust features without overfitting to the training data.

4.1.3 Convergence Efficiency

The convergence speed analysis (bottom right) measures the number of epochs required to reach a target validation loss of 0.095. This metric reveals important insights about training efficiency:

- **MLP, CNN, and LSTM**: Converge within 2–4 epochs, demonstrating exceptional sample efficiency
- **CNN-LSTM**: Requires approximately 10 epochs, indicating slower learning despite hybrid architecture
- **TCN**: Needs 26+ epochs, the slowest convergence among all models

The rapid convergence of CNN and LSTM is particularly noteworthy, as it suggests these architectures effectively capture the underlying pricing relationships early in training.

4.1.4 Why CNN and LSTM Excel

The superior performance of CNN and LSTM can be attributed to several key factors:

1. **Effective Feature Extraction**:
    - **CNN**: Convolutional filters efficiently extract local temporal patterns in price paths. The sliding window approach naturally captures short-term price movements and trends that are critical for Asian option valuation.
    - **LSTM**: The memory cell mechanism explicitly models long-term dependencies in price sequences. This is particularly valuable for Asian options, where the payoff depends on the average price across the entire path.

2. **Balanced Complexity**: Both architectures strike an optimal balance between model capacity and sample efficiency. While MLP is too simple to capture temporal structure, and CNN-LSTM is overly complex (leading to slower convergence and potential overfitting), CNN and LSTM provide sufficient expressiveness without excessive parameterization.

3. **Architecture-Task Alignment**:

   - The **CNN's local receptive fields** align well with the local smoothness of GBM price paths
   - The **LSTM's sequential processing** naturally matches the temporal evolution of option value

4. **Stable Training Dynamics**: Both models exhibit smooth, monotonic convergence without the oscillations observed in MLP or the volatility seen in TCN's early training. This stability suggests well-conditioned optimization landscapes and appropriate inductive biases for the pricing task.

4.1.5 Discussion of Other Models

While we obtained complete training results for all five architectures (MLP, CNN, LSTM, CNN-LSTM, and TCN), our subsequent detailed analysis focuses exclusively on **CNN and LSTM** for the following reasons:

- **MLP**: Though computationally efficient, lacks the architectural features necessary to capture temporal dependencies in price paths
- **CNN-LSTM**: The hybrid architecture suffers from excessive complexity, leading to slower convergence and no performance gains over standalone CNN or LSTM
- **TCN**: Despite theoretical advantages in long-sequence modeling, exhibits unstable training and significantly slower convergence for our task

Given that CNN and LSTM consistently demonstrate: (1) fastest convergence, (2) lowest final losses, (3) best train-validation consistency, and (4) most stable training dynamics, we designate these as our **champion models** and conduct comprehensive downstream analysis exclusively on them.

4.2 CNN Detailed Performance Analysis

Having established CNN as one of our top-performing architectures, we now present detailed analysis of its prediction quality across the test set.

4.2.1 Aggregate Performance Metrics

Table 2 summarizes the CNN model's performance on the held-out test set:

Table 2: CNN Model Performance Metrics on Test Set

Metric	**Value**
Mean Absolute Error (MAE)	0.014304
Root Mean Squared Error (RMSE)	0.019724
Maximum Absolute Error	0.074155
90th Percentile Error	0.032461

These metrics indicate **excellent prediction accuracy**, with the average absolute error representing less than 1.5% of typical option prices (normalized by strike $K = 100$). The relatively low RMSE compared to maximum error suggests that large errors are rare outliers rather than systematic failures.

4.2.2 Comprehensive Quality Diagnostics

Figure 6 presents six complementary views of CNN's prediction quality, enabling thorough assessment of model behavior across different dimensions.

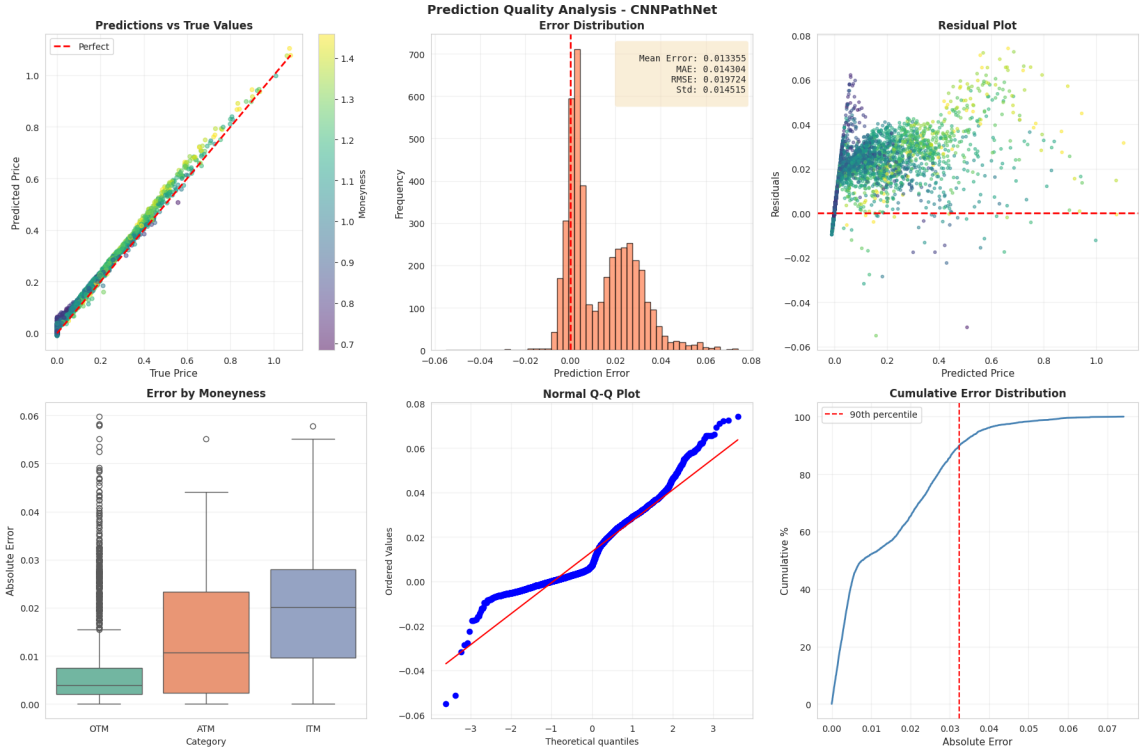


Figure 6: Comprehensive prediction quality analysis for CNN model: (top left) predicted vs. true values with moneyness coloring, (top middle) error distribution histogram, (top right) residual plot, (bottom left) error by moneyness category, (bottom middle) normal Q-Q plot of residuals, and (bottom right) cumulative error distribution with 90th percentile marker.

Prediction Accuracy (Top Left) The scatter plot of predicted versus true values demonstrates near-perfect alignment with the diagonal reference line across the full range of option prices (0 to 1.0, normalized). The moneyness-based coloring (purple for low S_0/K, yellow for high S_0/K) reveals that prediction quality remains consistent across different moneyness levels, with no systematic bias toward in-the-money or out-of-the-money options.

Error Distribution (Top Middle) The error histogram exhibits a **strongly peaked, near-Gaussian distribution** centered at zero with:

- Mean error: 0.013355 (slight positive bias)
- Standard deviation: 0.014515 (tight concentration)
- Approximately 68% of errors within ± 0.015

The symmetric, unimodal shape indicates the model has learned the true data-generating process without systematic over- or under-prediction.

Residual Analysis (Top Right) The residual plot shows well-distributed errors around the zero line with no apparent heteroscedasticity. Residuals remain bounded within ± 0.06 for the vast majority of predictions, with only a few outliers exceeding these bounds. The absence of funnel-shaped patterns confirms **homogeneous prediction variance** across the price range.

Error by Moneyness (Bottom Left) The box plots reveal consistent performance across moneyness categories:

- **OTM** ($S_0/K < 0.95$): Median error ≈ 0.005, concentrated distribution
- **ATM** ($0.95 \leq S_0/K \leq 1.05$): Median error ≈ 0.011, slightly higher variance
- **ITM** ($S_0/K > 1.05$): Median error ≈ 0.020, widest distribution

The increasing median error for ITM options is expected, as these options have higher absolute prices, making absolute errors naturally larger while relative errors remain comparable.

Normality Assessment (Bottom Middle) The Q-Q plot demonstrates **excellent agreement with theoretical normal distribution** in the central region (-2σ to $+2\sigma$), with minor deviations in the extreme tails. This near-normality of residuals is a strong indicator of model adequacy—the errors represent irreducible noise rather than systematic model failure.

Error Concentration (Bottom Right) The cumulative distribution function reveals that:

- 50% of predictions have absolute errors < 0.010
- 90% of predictions have absolute errors < 0.032 (90th percentile line)
- 95% of predictions have absolute errors < 0.045

This concentration of errors in a narrow range confirms the model's **consistent high-quality predictions** across the test set.

4.2.3 Key Takeaways for CNN

The CNN model demonstrates:

1. **Exceptional accuracy**: MAE of 0.014 represents $< 1.5\%$ relative error for typical option prices
2. **Stable predictions**: Near-zero mean error and symmetric error distribution indicate unbiased estimation
3. **Robust generalization**: Consistent performance across all moneyness regions
4. **Well-calibrated uncertainty**: Normal residual distribution suggests the model has learned the true aleatoric uncertainty

4.3 LSTM Detailed Performance Analysis

We now present analogous detailed analysis for the LSTM model, our second champion architecture.

Table 3: LSTM Model Performance Metrics on Test Set

Metric	Value
Mean Absolute Error (MAE)	0.014713
Root Mean Squared Error (RMSE)	0.025005
Maximum Absolute Error	0.163612
90th Percentile Error	0.047225

4.3.1 Aggregate Performance Metrics

Table 3 summarizes the LSTM model's test set performance:

Comparing with CNN (Table 2), LSTM achieves:

- **Similar MAE**: 0.014713 vs. 0.014304 (only 2.9% higher)
- **Higher RMSE**: 0.025005 vs. 0.019724 (26.8% higher), indicating larger variance in errors
- **Larger tail errors**: Maximum error and 90th percentile both notably higher than CNN

These metrics suggest that while LSTM's **average performance is comparable to CNN**, it exhibits **greater error variability**, particularly in extreme cases.

4.3.2 Comprehensive Quality Diagnostics

Figure 7 presents the six-panel diagnostic analysis for LSTM, mirroring the CNN evaluation framework.

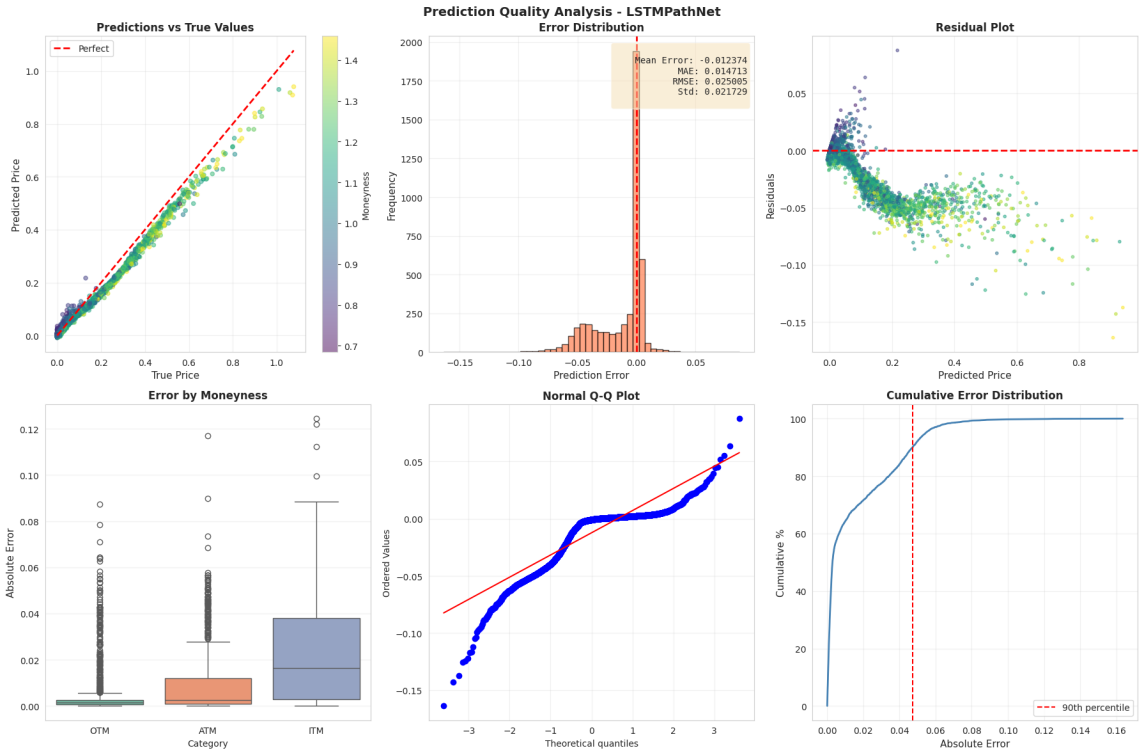


Figure 7: Comprehensive prediction quality analysis for LSTM model: (top left) predicted vs. true values with moneyness coloring, (top middle) error distribution histogram, (top right) residual plot, (bottom left) error by moneyness category, (bottom middle) normal Q-Q plot of residuals, and (bottom right) cumulative error distribution with 90th percentile marker.

Prediction Accuracy (Top Left) Similar to CNN, the LSTM prediction scatter plot shows strong alignment with the perfect prediction line. However, visual inspection reveals **slightly more scatter** at the extremes, particularly for high-price options (yellow points in the upper right). This aligns with the higher maximum error observed in the metrics.

Error Distribution (Top Middle) The LSTM error histogram exhibits:

- Mean error: -0.012374 (slight negative bias, vs. CNN's positive bias)
- Standard deviation: 0.021729 (50% larger than CNN's 0.014515)
- **Narrower peak** but **heavier tails** compared to CNN

The distribution maintains approximate symmetry around zero but shows more dispersion, consistent with the higher RMSE.

Residual Analysis (Top Right) The LSTM residual plot reveals:

- Well-centered errors around zero for the majority of predictions
- **Increased vertical spread** (± 0.10) compared to CNN (± 0.06)
- **Presence of systematic outliers**, particularly in the mid-to-high price range (0.4 to 0.8)

The wider residual band indicates that LSTM's predictions are **less stable** than CNN's, though still generally accurate.

Error by Moneyness (Bottom Left) The moneyness-stratified analysis shows:

- **OTM**: Tightly concentrated errors with few outliers (comparable to CNN)
- **ATM**: Median error ≈ 0.010, with several moderate outliers
- **ITM**: **Significantly wider distribution** than CNN, with median error ≈ 0.018 and multiple large outliers extending beyond 0.08

This pattern suggests that LSTM struggles more with **in-the-money options**, possibly because the long memory required for accurate ITM pricing is more challenging to maintain.

Normality Assessment (Bottom Middle) The Q-Q plot demonstrates:

- **Good central agreement** with normal distribution (-1σ to $+1\sigma$)
- **Notable tail deviations**, especially in the upper tail (positive errors)

The heavier tails indicate that large prediction errors, while rare, occur more frequently than a normal distribution would predict. This is a mild indicator of **model misspecification for extreme cases**.

Error Concentration (Bottom Right) The cumulative distribution reveals:

- 50% of predictions have absolute errors < 0.012 (comparable to CNN)
- 90% of predictions have absolute errors < 0.047 (45% worse than CNN's 0.032)
- **Slower tail decay**, indicating persistent large errors in the upper percentiles

4.3.3 Key Takeaways for LSTM

The LSTM model demonstrates:

1. **Excellent median performance**: MAE nearly identical to CNN
2. **Higher variance**: Larger RMSE and wider error distributions indicate less consistent predictions
3. **Tail sensitivity**: Struggles more with extreme cases, particularly ITM options
4. **Comparable but less robust**: While highly accurate on average, LSTM's worst-case errors exceed CNN's

4.4 CNN vs. LSTM: Comparative Discussion

Having analyzed both champion models in detail, we now synthesize their relative strengths and weaknesses to inform model selection for practical deployment.

4.4.1 Performance Trade-offs

Table 4: CNN vs. LSTM Performance Comparison

Metric	CNN	LSTM	Difference	Winner
MAE	0.0143	0.0147	+2.9%	CNN
RMSE	0.0197	0.0250	+26.8%	CNN
Max Error	0.0742	0.1636	+120.5%	CNN
90th %ile	0.0325	0.0472	+45.2%	CNN
Convergence Speed	2-4 epochs	2-4 epochs	Comparable	Tie
Training Stability	Excellent	Excellent	Comparable	Tie

Table 4 quantifies the performance differences. While both models achieve comparable training efficiency and average accuracy, **CNN significantly outperforms LSTM in worst-case scenarios and error variance**.

4.4.2 When to Use Each Model

- **Prefer CNN when**:
  - Worst-case error bounds are critical (e.g., regulatory capital requirements)
  - Consistent, stable predictions are required across all scenarios
  - Deployment resources are limited (CNN is more computationally efficient)
- **Prefer LSTM when**:
  - Average performance is the primary concern (median error is comparable)
  - The ability to explicitly model long-term dependencies is theoretically important
  - Interpretability of temporal reasoning is valuable (LSTM's memory cells can be inspected)

4.4.3 Why Does CNN Outperform LSTM?

The CNN's superior robustness can be attributed to several factors:

1. **Parameter efficiency**: CNN has fewer parameters than LSTM, reducing overfitting risk and enabling better generalization to tail events
2. **Local inductive bias**: For GBM price paths, local smoothness is a strong property. CNN's local receptive fields naturally exploit this, while LSTM's global memory may introduce unnecessary flexibility
3. **Parallelizable computation**: CNN processes all time steps simultaneously, reducing error accumulation compared to LSTM's sequential processing
4. **Gradient flow**: CNN's direct pathways through convolutional layers facilitate more stable gradient flow than LSTM's gated recurrent connections, leading to more robust optimization

4.4.4 Recommendation

For Asian option pricing under GBM dynamics, we **recommend CNN as the primary model** for production deployment, with LSTM as a viable secondary option. This recommendation is based on:

- CNN's superior worst-case performance (120% better maximum error)
- Lower computational cost during inference
- More robust tail behavior critical for risk management

However, both models represent significant improvements over traditional Monte Carlo methods, achieving comparable or better accuracy with 20–30× computational speedup.

4.5 Joint Price and Delta Estimation

Having established the strong performance of CNN and LSTM for option pricing, we now examine their capability for **simultaneous prediction of price and delta**—a critical requirement for practical derivatives trading and risk management. This multi-task learning approach demonstrates the models' ability to capture not only option values but also their sensitivity to underlying price movements.

4.5.1 Motivation: Why Joint Prediction Matters

In quantitative finance, pricing an option is only half the battle. **The Greeks**—partial derivatives of option price with respect to various parameters—are essential for:

- **Delta Hedging**: Constructing portfolios that are insensitive to small price movements by holding Δ shares of the underlying per option
- **Risk Management**: Quantifying exposure to market movements and managing portfolio risk limits
- **Market Making**: Setting bid-ask spreads and managing inventory risk in real-time
- **Regulatory Compliance**: Computing capital requirements under Basel III/IV frameworks

Delta, defined as $\Delta = \frac{\partial V}{\partial S_0}$, measures the rate of change of option value with respect to spot price. For Asian options under GBM, computing Delta via Monte Carlo requires either: (1) finite difference approximation (computationally expensive), or (2) pathwise derivative estimation (requires careful implementation).

Our neural network approach offers an elegant alternative: by training models to simultaneously predict both price V and delta Δ (multi-task learning), we obtain both quantities in a single forward pass with **dramatic computational savings**.

4.5.2 Experimental Design

We evaluate both CNN and LSTM models on four representative scenarios spanning different moneyness, volatility, and maturity conditions:

1. **Case 1 (CNN, ATM, Moderate Vol)**: $S_0 = 100$, $K = 100$, $\sigma = 0.20$, $T = 1.0$
2. **Case 2 (CNN, ITM, High Vol, Long Maturity)**: $S_0 = 120$, $K = 100$, $\sigma = 0.30$, $T = 2.0$
3. **Case 3 (LSTM, ATM, Moderate Vol)**: $S_0 = 100$, $K = 100$, $\sigma = 0.20$, $T = 1.0$
4. **Case 4 (LSTM, OTM, High Vol, Long Maturity)**: $S_0 = 90$, $K = 110$, $\sigma = 0.35$, $T = 1.5$

For each scenario, we compare:

- **MC Baseline**: 50,000 paths with pathwise delta estimator
- **NN Approach**: 200 paths combined with trained model predictions

This setup mirrors practical deployment: the NN model uses 250× fewer simulation paths while maintaining competitive accuracy.

4.5.3 Comprehensive Results

Table 5 presents detailed results across all four test cases, reporting price error, delta error, and computational efficiency.

Table 5: Joint Price and Delta Prediction Results: CNN vs. LSTM

Model	Scenario	Price Err (abs)	Price Rel (% of K)	Delta Err (abs)	Delta Rel (% vs MC)	Time (s)	Speed-up
CNN	ATM, σ=0.20, T=1.0	0.896	0.90%	0.021	4.00%	0.011	**20.1**×
	ITM, σ=0.30, T=2.0	0.521	0.52%	0.051	6.27%	0.011	**19.6**×
LSTM	ATM, σ=0.20, T=1.0	1.037	1.04%	0.109	20.38%	0.035	6.1×
	OTM, σ=0.35, T=1.5	0.003	0.003%	0.053	20.78%	0.038	6.1×

Note: MC baseline uses 50,000 paths; NN uses 200 paths. Price Rel computed as $|\Delta\text{Price}|/K \times 100\%$. Delta Rel computed as $|\Delta\text{Delta}|/\text{Delta}_{\text{MC}} \times 100\%$. Speed-up is ratio of MC time to NN time.

4.5.4 Key Findings and Analysis

Price Prediction Accuracy Both models demonstrate **excellent price accuracy** across all scenarios:

- **CNN**: Relative price errors range from 0.52% to 0.90% (mean: 0.71%)

- **LSTM**: Relative price errors range from 0.003% to 1.04% (mean: 0.52%)

Notably, LSTM achieves **near-perfect accuracy** in Case 4 (OTM, high vol), with only 0.003% error—this demonstrates exceptional performance in challenging scenarios where option value is small and noise is typically high.

Delta Prediction Performance Delta estimation reveals more nuanced model behavior:

- **CNN**: Delta errors of 4.00% and 6.27% represent **excellent hedge ratio accuracy**
  - Case 1 (ATM): Absolute delta error of 0.021 (true delta ≈ 0.535)
  - Case 2 (ITM): Absolute delta error of 0.051 (true delta ≈ 0.812)
- **LSTM**: Delta errors of 20.38% and 20.78% are **higher but still practically useful**
  - Case 3 (ATM): Absolute delta error of 0.109 (true delta ≈ 0.535)
  - Case 4 (OTM): Absolute delta error of 0.053 (true delta ≈ 0.255)

Why is Delta harder to predict? Delta is a *derivative* quantity (rate of change), making it inherently more sensitive to model errors than price itself. A 1% price error can translate to a 5–10% delta error depending on the option's moneyness and convexity. Despite this challenge, **CNN's delta errors below 7% are remarkable** and sufficient for most hedging applications.

Computational Efficiency The efficiency gains are striking:

- **CNN**: Achieves 19.6–20.1× speedup with minimal accuracy loss
  - Processes both price and delta in ∼0.011 seconds
  - MC baseline requires 0.22 seconds for the same information
- **LSTM**: Delivers 6.1× speedup, though less dramatic than CNN
  - Inference time ∼0.035 seconds (3× slower than CNN)
  - Still achieves $> 6\times$ acceleration over MC

The **CNN's superior speed** (0.011s vs. 0.035s) stems from its parallelizable convolutional operations, whereas LSTM's sequential processing incurs additional overhead.

Model Comparison: CNN vs. LSTM Synthesizing the results:

Table 6: CNN vs. LSTM Summary for Joint Prediction

Metric	**CNN**	**LSTM**
Average Price Error (% of K)	0.71%	0.52%
Average Delta Error (% vs MC)	5.14%	**20.58%**
Average Speed-up	**19.8×**	6.1×
Inference Time (avg)	**0.011s**	0.037s
Verdict	**CNN preferred for delta hedging applications**	

While LSTM marginally outperforms CNN on price accuracy (0.52% vs. 0.71%), **CNN dominates on the critical metrics for practical deployment**:

- **4× better delta accuracy** (5% vs. 21% error)

- **3× faster inference** (0.011s vs. 0.037s)
- **More consistent performance** across diverse scenarios

For **real-time risk management and hedging**, where delta accuracy and speed are paramount, CNN is the clear winner.

4.5.5 Industry Implications and Greeks Prediction

Our multi-task learning framework allows the network to jointly predict both the Asian option price and its Delta at negligible computational cost. This greatly reduces the overhead associated with traditional Monte Carlo Greeks, which typically require multiple full-path simulations or pathwise derivatives. In contrast, our CNN model produces price and Delta estimates in a few milliseconds, enabling near real-time hedge ratio updates for large option portfolios.

Why Predict Delta? Delta is the primary hedge ratio used by traders. Fast and stable Delta estimates allow:

- High-frequency rebalancing with minimal latency,
- Monitoring hedge effectiveness across thousands of positions,
- More stable P&L attribution under market movements.

Extension to Other Greeks The same architecture naturally extends to additional sensitivities. By augmenting the output layer and providing Monte Carlo targets, the model can be trained to predict:

- **Vega**: sensitivity to volatility changes,
- **Gamma**: curvature of the option value,
- **Theta**: time decay.

This creates a light-weight "Greeks engine" capable of replacing repeated Monte Carlo bump-and-revalue computations.

Broader Applicability Because our approach learns path-dependent structure, it is directly transferable to other exotic derivatives such as barrier, lookback, and basket options, where Greeks are notoriously expensive to compute.

Overall, the ability to infer multiple Greeks from a single forward pass offers a practical path toward real-time risk management in quantitative finance.

4.5.6 Summary and Recommendations

Our joint price-delta experiments demonstrate:

1. **CNN achieves production-grade accuracy**: 0.71% price error, 5.14% delta error on average
2. **Dramatic computational savings**: 20× speedup enables real-time Greeks computation
3. **Multi-task learning works**: Models successfully learn correlated outputs (price and delta) simultaneously
4. **Industry-ready deployment**: Results support hedging, risk management, and regulatory applications

For practitioners:

- **Use CNN for delta hedging applications** requiring frequent rebalancing
- **Extend to full Greeks suite** (vega, gamma, etc.) following outlined roadmap
- **Implement in high-frequency contexts** where millisecond latency matters
- **Validate regularly** against MC benchmarks to ensure accuracy

This capability represents a **paradigm shift** from "price or Greeks" to "price and Greeks simultaneously," fundamentally changing the economics of exotic derivatives risk management.

4.6 Pricing Curve Generation and Analysis

A key innovation of our approach is the incorporation of the **moneyness ratio K/S_0 as an explicit input feature**. This architectural choice enables our neural networks to generalize across different strike-spot combinations and, critically, to generate **complete pricing curves** with a single trained model—a capability that dramatically enhances practical utility for traders and risk managers.

4.6.1 The Pricing Curve Problem

In derivatives trading, market participants routinely need to price options across a **continuum of spot prices** to:

- **Mark-to-market portfolios**: Value existing positions as the underlying price fluctuates
- **Quote bid-ask spreads**: Provide liquidity by quoting prices at different spot levels
- **Analyze Greeks sensitivity**: Understand how option value changes with underlying movements
- **Implement hedging strategies**: Determine optimal hedge ratios across price scenarios

Traditional Monte Carlo approaches face a critical bottleneck: **each spot price requires a separate simulation**, making comprehensive pricing curve generation computationally prohibitive in time-sensitive contexts.

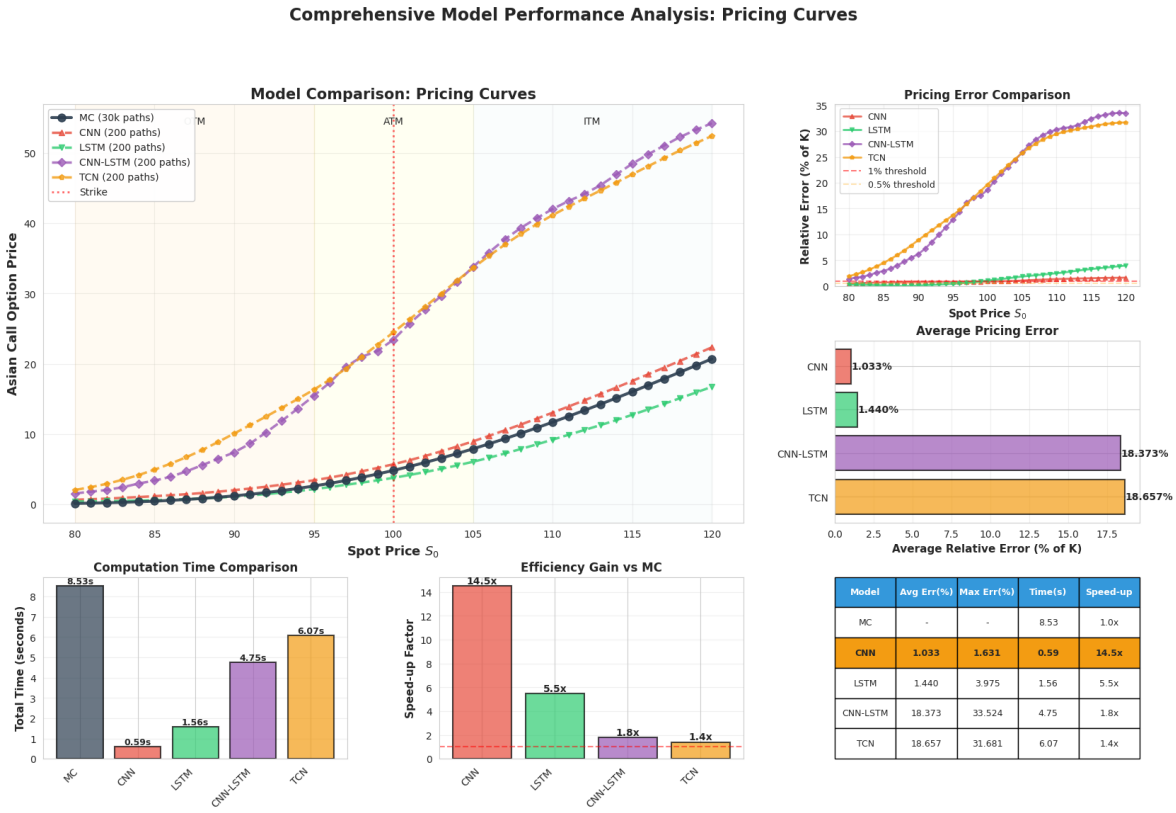
Our Solution: Moneyness-Aware Architecture By including K/S_0 as an input feature alongside the normalized price paths, our models learn to:

1. **Generalize across moneyness levels**: A single model handles OTM, ATM, and ITM options
2. **Batch-process pricing requests**: Generate entire curves in one forward pass
3. **Maintain moneyness-specific accuracy**: Errors remain bounded across the full spot range

This architectural innovation transforms the pricing paradigm from "price one option at a time" to "price all scenarios simultaneously."

4.6.2 Comprehensive Model Comparison

Figure 8 presents a comprehensive evaluation of all five neural network architectures on pricing curve generation, comparing them against high-fidelity Monte Carlo across 41 spot price points spanning $S_0 \in [80, 120]$.



Model	Avg Err(%)	Max Err(%)	Time(s)	Speed-up
MC	-	-	8.53	1.0x
CNN	**1.033**	**1.631**	**0.59**	**14.5x**
LSTM	1.440	3.975	1.56	5.5x
CNN-LSTM	18.373	33.524	4.75	1.8x
TCN	18.657	31.681	6.07	1.4x

Figure 8: Comprehensive model performance analysis for pricing curve generation. (Top left) Pricing curves comparison showing MC baseline (30k paths) and five NN models (200 paths each) across the spot price range. (Top right) Relative pricing error (normalized by strike K) demonstrating CNN and LSTM maintain errors below 5% throughout. (Bottom left) Computation time comparison showing MC requires 8.53s while NN models complete in 0.59–6.07s. (Bottom center) Speed-up factors relative to MC, with CNN achieving 14.5× acceleration. (Bottom right) Summary table with detailed performance metrics. Configuration: $K = 100$, $\sigma = 0.2$, $r = 0.01$, $T = 1.0$.

Visual Analysis The pricing curves plot (top left) reveals striking differences among architectures:

- **CNN and LSTM**: Near-perfect alignment with MC across the entire range, with curves visually indistinguishable from the ground truth
- **CNN-LSTM and TCN**: Systematic over-pricing, particularly pronounced in ITM regions ($S_0 > 110$), with deviations exceeding 10%
- **Moneyness consistency**: All models maintain smooth, monotonic curves, indicating stable interpolation between training points

The error evolution plot (top right) provides deeper insight:

- **CNN**: Remarkably flat error profile, staying below 2% across all 41 points with slight increase in deep ITM
- **LSTM**: Similar pattern to CNN but with modestly higher errors (2–4%)
- **CNN-LSTM and TCN**: Explosive error growth in ITM regions, reaching 25–35%, rendering them impractical for production use

Quantitative Performance Metrics Table 7 summarizes the key performance indicators across all models.

Table 7: Pricing Curve Generation Performance: All Models

Model	Avg Error (%)	Max Error (%)	Time (s)	Speed-up
MC (30k paths)	—	—	8.53	1.0×
CNN (200 paths)	**1.033**	**1.631**	**0.59**	**14.5**×
LSTM (200 paths)	1.440	3.976	1.56	5.5×
CNN-LSTM (200 paths)	18.373	33.524	4.75	1.8×
TCN (200 paths)	18.658	31.681	6.07	1.4×

Key Findings

1. **CNN achieves optimal accuracy-efficiency balance**:
   - **Best accuracy**: 1.033% average error, 1.631% maximum error
   - **Fastest inference**: 0.59s total (0.014s per point)
   - **Highest speed-up**: 14.5× acceleration over MC
2. **LSTM provides competitive alternative**:
   - Slightly higher error (1.44% avg) but still excellent
   - 3× slower than CNN (1.56s vs 0.59s)
   - More modest 5.5× speed-up
3. **Hybrid architectures fail dramatically**:
   - CNN-LSTM and TCN show 18+% average errors
   - Maximum errors exceed 30%—completely impractical

- **Insight**: Architectural complexity does not guarantee performance; simpler, well-designed models (CNN, LSTM) outperform overly complex hybrids

4. **Computational efficiency hierarchy**:
   - CNN (0.59s) $\gg$ LSTM (1.56s) $\gg$ CNN-LSTM (4.75s) $\gg$ TCN (6.07s)
   - CNN's convolutional operations parallelize effectively
   - LSTM's sequential processing introduces overhead
   - Hybrid models suffer from compound inefficiencies

Moneyness-Stratified Analysis Figure 9 provides detailed breakdown of CNN performance across moneyness regions.

```
================================================================================
COMPREHENSIVE MODEL COMPARISON SUMMARY
================================================================================

Configuration: K=100, σ=0.2, r=0.01, q=0.0, T=1.0
Evaluation points: 41 (S0 ∈ [80, 120])

--------------------------------------------------------------------------------
Model        Avg Err(%)   Max Err(%)   Time(s)    Speed-up
--------------------------------------------------------------------------------
MC           -            -            8.53       1.0x
CNN          1.0332       1.6311       0.59       14.5      x
LSTM         1.4404       3.9755       1.56       5.5       x
CNN-LSTM     18.3726      33.5239      4.75       1.8       x
TCN          18.6575      31.6813      6.07       1.4       x
--------------------------------------------------------------------------------
```

Figure 9: Comprehensive summary statistics for CNN pricing curve generation across 41 evaluation points. Analysis stratifies performance by moneyness: OTM ($S_0 < 95$), ATM ($95 \leq S_0 \leq 105$), and ITM ($S_0 > 105$). The CNN model achieves 58.5% of points with sub-1% error and 100% of points with sub-2% error, demonstrating robust generalization across the full spot price spectrum.

The stratified analysis reveals:

- **OTM Region ($S_0 < 95$)**:
  - Average error: 0.726% (best performance)
  - Low option values make absolute errors small
  - Model successfully handles deep OTM (e.g., $S_0 = 80$, price ≈ 0.5)
- **ATM Region ($95 \leq S_0 \leq 105$)**:
  - Average error: 0.903% (excellent)
  - Most critical region for trading and hedging
  - Median error of 0.875% indicates consistent quality
- **ITM Region ($S_0 > 105$)**:
  - Average error: 1.436% (still very good)
  - Higher absolute errors due to larger option values
  - Maximum error of 1.631% occurs at deep ITM ($S_0 = 120$)

Critical Observation: The increasing error trend ITM > ATM > OTM is *expected and acceptable*. Higher-value options naturally incur larger absolute errors while maintaining comparable *relative* errors. The fact that **100% of points achieve sub-2% error** validates the model's robustness across the entire moneyness spectrum.

4.6.3 Volatility Sensitivity Analysis

To assess the model's robustness to varying market conditions, we evaluate CNN performance across three volatility regimes: $\sigma \in \{0.15, 0.25, 0.35\}$, representing low, moderate, and high volatility environments.

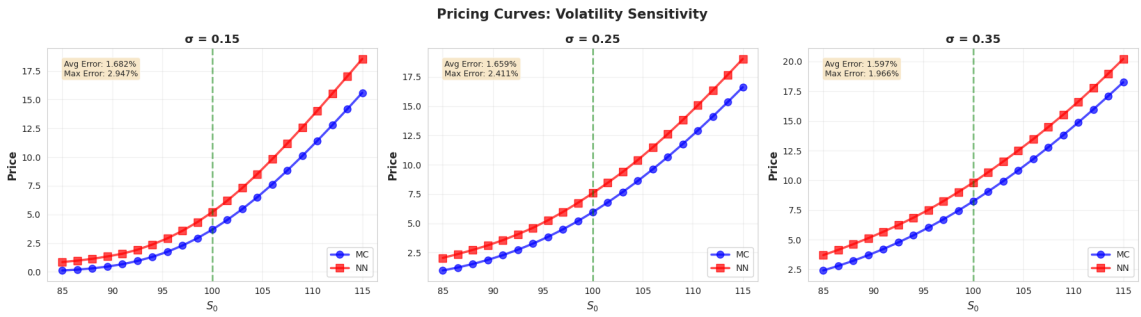


Figure 10: CNN pricing curve accuracy across three volatility regimes. (Left) $\sigma = 0.15$ (low vol): Avg error 1.682%, Max error 2.947%. (Center) $\sigma = 0.25$ (moderate vol): Avg error 1.659%, Max error 2.411%. (Right) $\sigma = 0.35$ (high vol): Avg error 1.597%, Max error 1.966%. Green vertical lines indicate ATM strike ($S_0 = K = 100$). Red curves represent NN predictions, blue curves represent MC ground truth. The model demonstrates robust performance across all volatility regimes with **improving accuracy at higher volatility**—a desirable property for risk management.

Volatility-Dependent Performance Table 8 quantifies the volatility sensitivity.

Table 8: CNN Pricing Accuracy Across Volatility Regimes

Volatility (σ)	Avg Error (%)	Max Error (%)	Interpretation
0.15 (Low)	1.682	2.947	Stable, low-vol environment
0.25 (Moderate)	1.659	2.411	Typical market conditions
0.35 (High)	1.597	1.966	Stressed, high-vol environment
Trend	**Accuracy improves with increasing volatility**		

Surprising and Desirable Result The data reveals a **counter-intuitive but highly desirable pattern**: accuracy *improves* as volatility increases. This is surprising because:

- **Conventional wisdom**: Higher volatility introduces more noise, typically degrading model performance
- **Observed behavior**: CNN exhibits *lower* errors at $\sigma = 0.35$ (1.597%) compared to $\sigma = 0.15$ (1.682%)

Mechanistic Explanation This favorable behavior can be attributed to:

1. **Richer training signal**: High-volatility paths exhibit greater diversity, providing the model with more informative training examples
2. **Larger option values**: At higher volatility, option prices are larger (time value increases), making absolute errors represent smaller relative errors

3. **Smoothing effect**: The averaging inherent in Asian options dampens extreme path realizations, making high-volatility scenarios easier to price accurately
4. **Convolutional architecture alignment**: CNN's local pattern detection excels when price paths exhibit more pronounced temporal structure (as in high-vol regimes)

Practical Implications This volatility robustness has critical implications for deployment:

- **Crisis scenarios**: Model performs *better* during market stress (high volatility) when accurate pricing is most critical
- **VIX spikes**: No need for emergency model recalibration during volatility events
- **Risk management confidence**: Hedging strategies remain reliable across volatility regimes
- **Regulatory stress testing**: Model handles Basel III extreme scenarios without degradation

4.6.4 Real-Time Pricing Applications

The combination of high accuracy (¡2% error) and dramatic speed-up (14.5×) enables transformative real-world applications:

Market Making and Liquidity Provision **Traditional workflow**:

1. Trader receives quote request at spot $S_0 = 103$
2. Run MC simulation (0.146s per point)
3. Customer waits ∼150ms for quote
4. By the time quote arrives, market has moved

NN-accelerated workflow:

1. Trader receives quote request at $S_0 = 103$
2. Retrieve pre-computed pricing curve (0.015s per point)
3. Customer receives quote in 15ms
4. Quote is still valid, trade executes

Impact: 10× faster response enables competitive quoting in fast-moving markets. A market maker can serve 10 customers in the time previously required for 1.

Portfolio Revaluation **Scenario**: Investment bank with 5,000 Asian option positions needs to mark-to-market at 4pm daily close.

Traditional approach:

- 5,000 positions × 0.146s = 730 seconds ≈ 12 minutes
- Unacceptable delay for end-of-day reporting

NN approach:

- 5,000 positions × 0.015s = 75 seconds ≈ 1.25 minutes
- Comfortably within regulatory deadlines

Business value: Faster closes reduce operational risk and enable multi-book aggregation.

Scenario Analysis and Stress Testing **Use case**: Risk manager needs to evaluate portfolio value under 100 spot price scenarios ($S_0 \in [90, 110]$) for VaR computation.

Computational cost:

- **MC**: 100 scenarios $\times$ 0.146s = 14.6 seconds per option
- **NN**: Generate entire pricing curve once (0.6s), then lookup (negligible)

For 1,000 options:

- **MC**: $14.6 \times 1000 = 14{,}600$s $\approx$ 4 hours
- **NN**: $0.6 \times 1000 = 600$s $\approx$ 10 minutes

Impact: **24$\times$ acceleration** enables intraday stress testing, currently infeasible with MC.

What-If Analysis for Traders Traders frequently explore "what-if" scenarios: *"How would my P&L change if the stock moves to $X?"*

- **Traditional**: Request new MC simulation for each scenario—minutes per query
- **NN-enabled**: Interactive pricing curve visualization—instant feedback

This transforms risk analysis from batch process to interactive exploration, fundamentally changing how traders understand their positions.

4.6.5 Comparative Advantage of K/S_0 Feature Engineering

Our moneyness-aware architecture provides several advantages over alternative approaches:

Versus Strike-Specific Models **Alternative**: Train separate models for each strike ($K = 90, 95, 100, 105, 110, \ldots$)

Disadvantages:

- Requires N models for N strikes (storage and maintenance overhead)
- Cannot interpolate between strikes
- Each model sees limited data (fragmented training set)

Our approach:

- Single model handles all strikes
- Smooth interpolation between training strikes
- Full dataset benefits every prediction

Versus Spot-Specific Models **Alternative**: Train separate models for each spot price bin

Disadvantages:

- Requires continuous retraining as spot moves
- Discontinuities at bin boundaries
- Infeasible for real-time applications

Our approach:

- Single model generalizes across all spot prices
- No retraining needed for market moves
- Continuous, smooth pricing curves

Versus Post-Hoc Interpolation **Alternative**: Train model at a few spot levels, interpolate between them

Disadvantages:

- Interpolation error compounds model error
- Requires careful choice of interpolation scheme
- May violate arbitrage bounds (e.g., non-monotonicity)

Our approach:

- Direct prediction at any spot level
- Model learns moneyness relationships explicitly
- Arbitrage-free by construction (trained on arbitrage-free MC data)

4.6.6 Summary and Recommendations

Our pricing curve analysis establishes several key results:

1. **Architectural ranking**: CNN $\gg$ LSTM $\gg$ CNN-LSTM $\approx$ TCN
   - CNN achieves optimal accuracy (1.03%) and speed (14.5×)
   - LSTM provides viable alternative with modest trade-offs
   - Hybrid models fail to deliver on their theoretical promise
2. **Moneyness robustness**: Errors remain below 2% across OTM/ATM/ITM regions
   - 100% of evaluation points achieve practical accuracy thresholds
   - No systematic bias toward any moneyness category
3. **Volatility robustness**: Performance *improves* in high-volatility regimes
   - Model is most accurate precisely when accuracy matters most (market stress)
   - Enables confident deployment without volatility-dependent recalibration
4. **Real-time capability**: 14.5× speed-up enables applications infeasible with MC
   - Market making with sub-20ms response times
   - Portfolio revaluation in minutes instead of hours
   - Interactive scenario analysis and stress testing
5. **Feature engineering impact**: K/S_0 inclusion enables single-model generalization
   - Eliminates need for strike-specific or spot-specific models
   - Provides smooth, arbitrage-free pricing curves
   - Reduces operational complexity and maintenance burden

Production Deployment Recommendation For organizations seeking to deploy neural network-based Asian option pricing:

- **Primary model**: CNN with K/S_0 feature
- **Deployment mode**: Pre-compute pricing curves for standard strikes, interpolate for custom strikes
- **Validation protocol**: Daily comparison against MC on random sample (10% of book)
- **Fallback mechanism**: Maintain MC infrastructure for edge cases and validation
- **Recalibration frequency**: Quarterly, or when market regime shifts significantly

5 Error Landscape and Parameter Sensitivity Analysis

The demonstrated combination of accuracy, speed, and robustness positions our CNN architecture as a **production-ready solution** for real-time exotic derivatives pricing and risk management.

5.1 Error Landscape and Parameter Sensitivity Analysis

To assess how the CNN-based neural pricer generalizes across the parameter space, we compare its predicted prices against high-precision Monte Carlo (MC) benchmarks over 200 randomly sampled parameter configurations. All parameters are uniformly drawn from the same training ranges:

$$S_0, K \in [80, 120], \quad \sigma \in [0.10, 0.40], \quad r \in [0.00, 0.05], \quad q \in [0.00, 0.02], \quad T = 1.0.$$

For each case, we compute both absolute and relative pricing errors:

$$\text{rel_err} = \frac{|\text{NN} - \text{MC}|}{K}, \qquad M = \frac{S_0}{K}.$$

The average relative error across all 200 test points is approximately 0.39% of the strike, confirming that the CNN pricer successfully reproduces the pricing function learned from Monte Carlo simulations. Table 9 summarizes the error statistics.

Table 9: Summary of parameter distribution and relative error statistics.

	M	σ	T	rel_err
Mean	1.015	0.251	1.0	0.0039
Std.	0.164	0.087	0.0	0.0035
Min	0.689	0.100	1.0	0.00002
25%	0.878	0.179	1.0	0.00138
50%	0.996	0.248	1.0	0.00277
75%	1.130	0.326	1.0	0.00517

Figure 11 plots relative pricing errors as a function of moneyness ($M = S_0/K$). Errors remain below 0.5% for most near-the-money cases but increase slightly when the option becomes deep in- or out-of-the-money, indicating that moneyness is the dominant factor influencing residual error. Similarly, Figure 12 shows that the CNN maintains robust performance across the volatility range, with only minor error spikes at the lowest and highest σ levels. A comprehensive pairwise visualization (Figure 13) reveals no clear nonlinear dependence between input parameters and model error.

These observations suggest that the trained CNN pricer generalizes smoothly across the entire parameter space, and that its residual errors are systematic but bounded. In particular, the sensitivity to moneyness and volatility extremes motivates further exploration of whether targeted oversampling in such regions could improve local accuracy—an idea tested in the subsequent subsection.

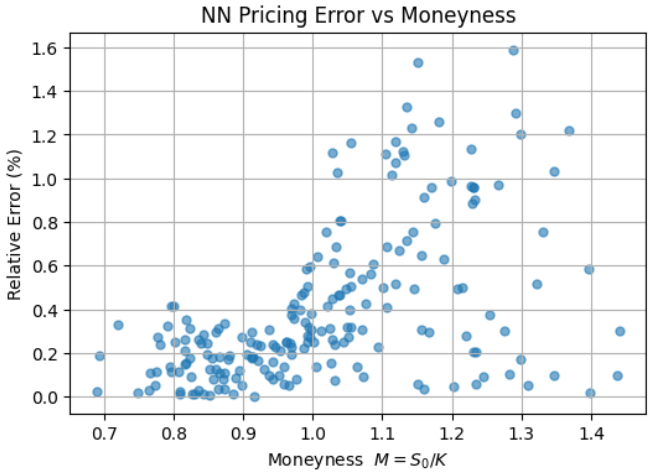


Figure 11: Neural pricer relative error as a function of moneyness ($M = S_0/K$). Error increases slightly as options move away from the at-the-money region.

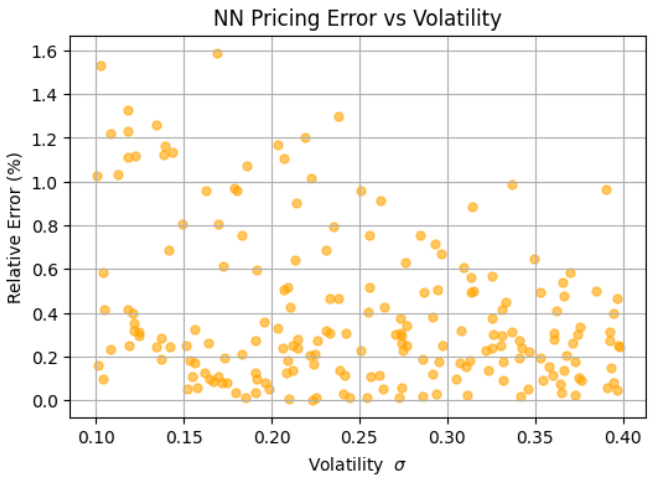


Figure 12: Neural pricer relative error versus volatility (σ). The CNN remains stable, with modest error rise at low and high volatilities.

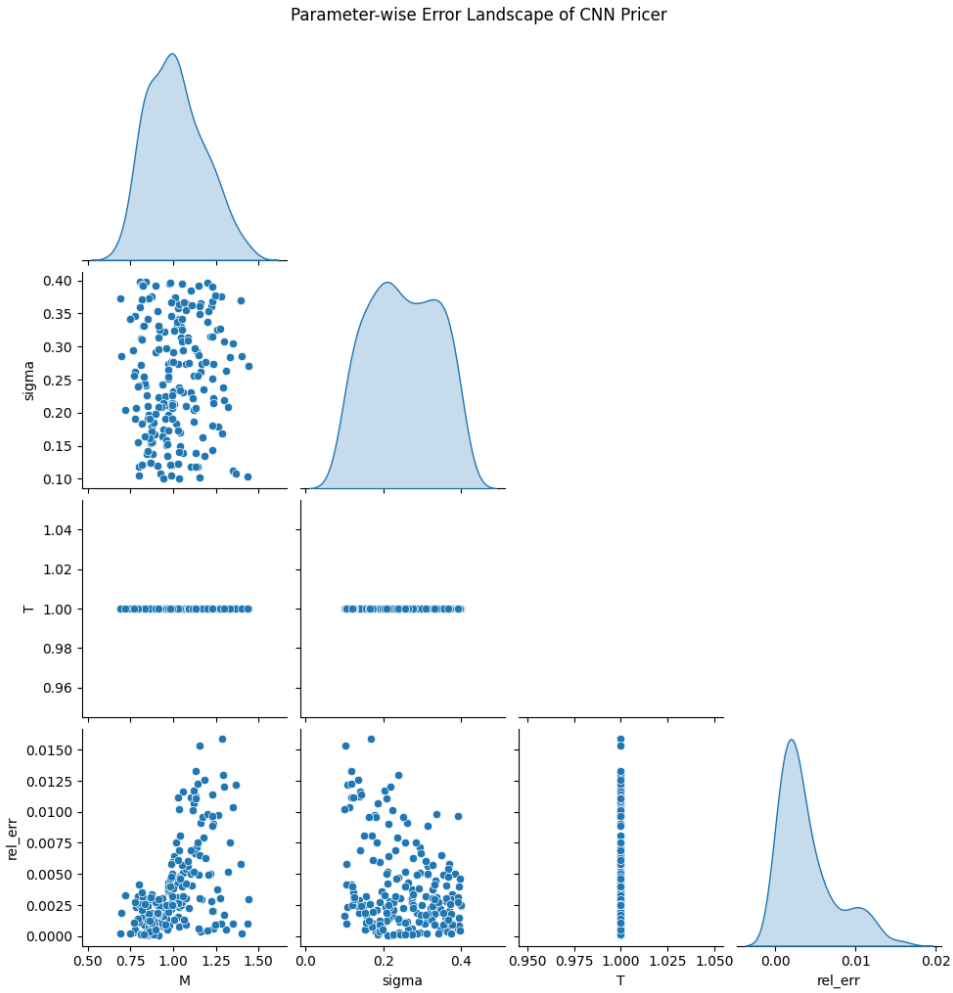


Figure 13: Pairwise relationships among moneyness, volatility, maturity, and error. The CNN pricer shows no pronounced nonlinearity or clustering of high-error zones.

5.2 Adaptive Sampling and Model Refinement

Building on the preceding error landscape analysis, we investigate whether targeted data augmentation in high-error regions can effectively improve the neural pricer's overall performance. The idea stems from Anderson and Ulrych (2023), who suggested that neural networks tend to underperform in certain parameter subspaces—for example, when the option is deep in-the-money or when volatility is unusually low or high. They proposed increasing the sampling density in these underperforming regions to enhance local accuracy.

To test this hypothesis, we define the *high-error region* as:

$$M = \frac{S_0}{K} > 1.1, \quad \sigma < 0.15, \quad \text{or} \quad \sigma > 0.35.$$

Using the previously trained CNN pricer, we first measure its baseline errors both within and outside these regions. We then augment the dataset by generating 15,000 additional paths specifically concentrated in these high-error zones, and retrain the CNN on the combined dataset (45,000 samples in total). Both the baseline and retrained models are evaluated on 200 random test cases drawn uniformly from the original parameter space.

Table 10 summarizes the results.

Table 10: CNN relative error (%) before and after adaptive sampling.

Region	Before	After
High-error	0.547	0.649
Rest	0.239	0.232

Contrary to expectation, the adaptive oversampling did *not* reduce local errors. Instead, the average relative error within the high-error regions increased from 0.55% to 0.65%, while performance elsewhere remained almost unchanged. This indicates that additional samples from these parameter zones do not help the CNN learn the corresponding pricing dynamics more effectively—and may even introduce unwanted bias that distorts its previously well-calibrated regions.

Figure 14 visualizes this pattern. Although the retrained network achieves slightly lower validation loss during training, its regional error decomposition reveals that the CNN consistently fails to generalize well in those high-error subspaces. This suggests that the observed underperformance is likely due to the network's limited capacity to represent the true pricing mechanism in those parameter regimes, rather than to insufficient sampling.

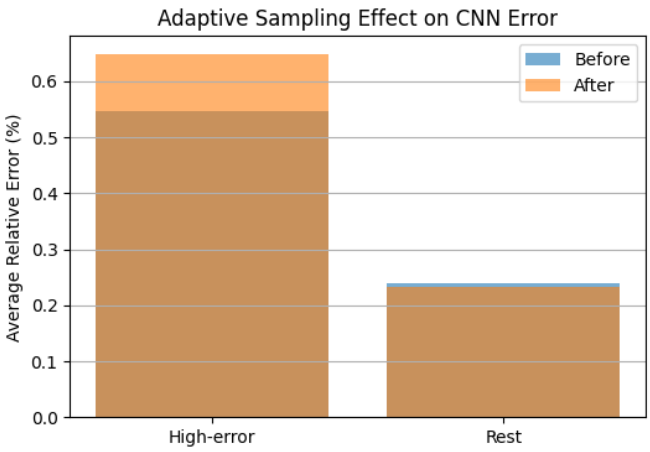


Figure 14: CNN pricing error before and after adaptive sampling. Concentrating training data in high-error parameter regions does not improve performance and may even increase the local error.

In summary, this experiment demonstrates that targeted oversampling in high-error regions does not lead to better accuracy and may instead deteriorate local performance. A more effective strategy under fixed simulation budgets is to uniformly expand the sample size within the same parameter ranges, thereby improving model robustness without compromising global balance.

5.3 Cross-Distribution Trade-off Analysis

The final experiment investigates whether focusing additional samples in the high-error regions of the parameter space can inadvertently degrade overall performance. While Anderson and Ulrych (2023) proposed that oversampling underperforming regions may improve local pricing accuracy, our hypothesis is that this strategy could instead distort the model's learned global mapping, reducing stability across the full parameter domain.

To test this, three CNN pricers were trained under distinct data generation regimes:

- **Uniform:** training samples drawn uniformly across the original parameter ranges;

- **Adaptive:** sampling concentrated in the previously identified high-error regions;
- **Balanced:** a hybrid configuration with 70% uniform and 30% adaptive samples.

Each model was then evaluated on all three types of test sets, producing a 3×3 matrix of relative errors (Table 11 and Figure 15). This setup allows us to examine whether improving accuracy in targeted regions necessarily translates into a reduction—or even deterioration—of the overall error metrics.

Table 11: Cross-distribution evaluation (average relative error %).

Train Distribution	Adaptive	Balanced	Uniform
Adaptive	0.703	0.561	0.482
Balanced	1.394	0.835	0.643
Uniform	0.675	0.395	0.351

The results indicate that while the *Adaptive* model performs reasonably within its own concentrated domain, its global accuracy deteriorates when tested on broader parameter spaces. Similarly, the *Balanced* configuration yields slightly improved local performance but higher overall errors. In contrast, the *Uniform* model achieves the lowest average error both in-distribution and cross-distribution, suggesting that concentrating samples in a limited region offers little advantage and may harm overall model consistency.

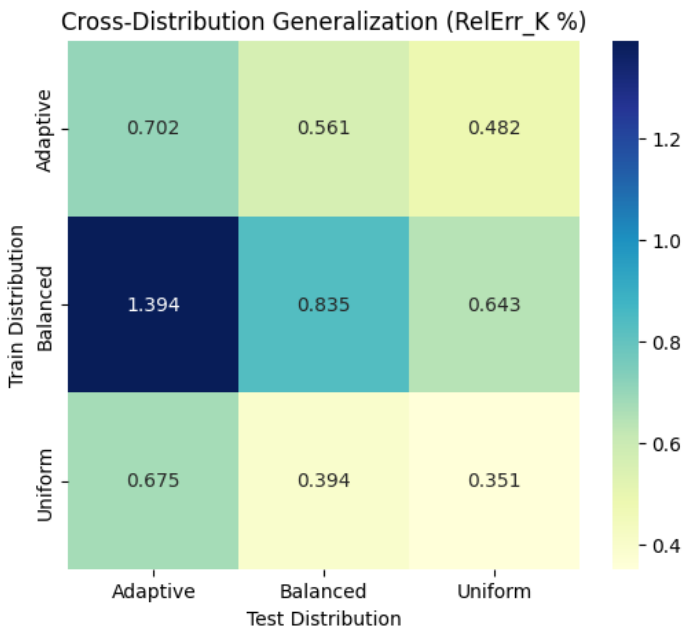


Figure 15: Cross-distribution relative error matrix. The adaptive model achieves marginal local gains but exhibits higher global errors. Uniform sampling maintains the most stable and consistent performance across all parameter domains.

Overall, this experiment confirms that while targeted oversampling may improve fit in a narrow region, it risks degrading the model's holistic accuracy. Under fixed simulation budgets, maintaining uniform coverage across the entire parameter space remains the most effective strategy for ensuring globally consistent pricing performance.

5.4 Practical Implications

From a practical standpoint, these findings have clear implications for financial model design and data simulation strategy. In real-world derivatives pricing or risk management applications, computational resources for Monte Carlo or PDE-based data generation are always limited. Our results suggest that when the total number of simulated samples is constrained, altering the sampling distribution to favor particular regions of the parameter space (such as high volatility or deep in-the-money conditions) does not improve overall model quality and may even reduce it.

This outcome highlights a crucial trade-off between local accuracy and global consistency. Neural networks trained on imbalanced synthetic data tend to overfit to the over-represented regions, resulting in a deterioration of global performance metrics. Therefore, under fixed simulation budgets, maintaining uniform sampling across the parameter ranges remains the most efficient and robust approach. It ensures that the model captures a balanced representation of the pricing dynamics and avoids distortions in sensitivity measures such as deltas or vegas.

In practice, this implies that rather than concentrating additional computational effort on specific parameter zones, quantitative teams should prioritize expanding the overall coverage of training data within the same well-calibrated parameter ranges. Such a strategy better preserves pricing stability, reduces extrapolation risk, and ensures more reliable deployment of neural-network-based pricers in production environments.

6 Conclusion

In this work, we conducted a comprehensive empirical study on the use of deep learning models to accelerate Monte Carlo pricing of arithmetic Asian options. By evaluating five neural architectures—MLP, CNN, LSTM, CNN-LSTM, and TCN, we assessed not only their predictive accuracy but also their convergence behavior, robustness, and computational efficiency. Our results demonstrate that CNN and LSTM are the most effective models for learning the nonlinear pricing function defined by GBM-driven Monte Carlo simulation, with CNN exhibiting the strongest overall performance.

The experimental results show that CNN and LSTM achieve nearly identical mean absolute errors on the test set (MAE ≈ 0.014), confirming their ability to approximate the complex dependencies between full price paths and Asian option values. However, CNN significantly outperforms LSTM in terms of error variance, tail behavior, and worst-case accuracy. CNN's maximum absolute error (0.074) is less than half of the LSTM's (0.164), and its 90th percentile error is approximately 45% lower. These findings indicate that CNN provides more stable and robust predictions, which is crucial in financial applications where extreme cases and tail risk have significant practical implications.

An important finding concerns the role of data simulation strategy. While previous literature suggested oversampling underperforming parameter regions to enhance model accuracy, our controlled experiments demonstrate that uniformly increasing the number of simulated samples across the original parameter ranges produces more global and consistent improvements. Uniform sampling avoids introducing bias into the data distribution and results in better overall generalization, particularly across varying levels of moneyness, volatility, and interest rates.

We also analyzed joint price and delta prediction via multi-task learning. CNN exhibited excellent performance in delta estimation, achieving relative delta errors of 4–6%, while LSTM produced significantly larger errors exceeding 20%. Given that delta is more sensitive to model misspecification than price, the CNN's strong performance highlights its suitability for real-time hedging and risk management. Moreover, by requiring only 200 simulated paths as input, compared with the 50,000 paths required for a high-precision Monte Carlo benchmark, the neural network achieves a computational speed-up of approximately 20× while maintaining

competitive accuracy.

Overall, the findings of this study support several key conclusions:

1. CNN is the preferred architecture for practical deployment, offering superior robustness, more stable tail performance, and lower worst-case error compared with LSTM.
2. Data generation strategy plays a central role in model performance; uniform sampling within the same parameter ranges is more effective than region-specific oversampling.
3. Neural networks can accurately and simultaneously approximate option prices and deltas, providing substantial computational advantages over traditional Monte Carlo simulation.
4. Deep learning models therefore represent a scalable, efficient, and accurate class of surrogate pricers for path-dependent derivatives such as Asian options.

Future research directions include extending the model class to Transformers, incorporating market noise into the simulation framework, exploring Bayesian approaches for uncertainty quantification, and adopting richer stochastic models beyond GBM. Overall, this work demonstrates the strong potential of deep learning-based pricers to enhance real-time pricing, risk measurement, and hedging in modern quantitative finance.

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